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A recent review of solution approaches for green vehicle routing problem and its variants

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ABSTRACT

The green vehicle routing problem (GVRP) has been a prominent topic in the literature on logistics and transportation, leading to extensive research and previous review studies covering various aspects. Operations research has seen the development of various exact and approximation approaches for different extensions of the GVRP. This paper presents an up-to-date and thorough review of GVRP literature spanning from 2016 to 2023, encompassing 458 papers. significant contribution lies in the updated solution approaches and algorithms applied to both single-objective and multi-objective GVRP. Notably, 92.58 % of the papers introduced a mathematical model for GVRP, with many researchers adopting mixed integer linear programming as the preferred modeling approach. The findings indicate that both metaheuristics and hybrid are the most employed solution approaches for addressing single-objective GVRP. Among hybrid approaches, the combination of metaheuristics-metaheuristics is particularly favored by GVRP researchers. Furthermore, large neighborhood search (LNS) and its variants (especially adaptive large neighborhood search) emerges as the most widely adopted algorithm in single-objective GVRP. These algorithms are proposed within both metaheuristic and hybrid approaches, where A-/LNS is often combined with other algorithms. Conversely, metaheuristics are predominant in addressing multi-objective GVRP, with NSGA-II being the most frequently proposed algorithm. Researchers frequently utilize GAMS and CPLEX as optimization modeling software and solvers. Furthermore, MATLAB is a commonly employed programming language for implementing proposed algorithms.

1. Introduction

Green logistics is a comprehensive field that addresses economic, environmental, and social aspects, aiming to minimize negative impacts on the environment while enhancing society and economic well-being. It focuses on actions that have a minimal negative impact on the environment and introduces tools and behaviors that improve society and its economic level improve. The integration of environmental considerations into logistics activities is crucial for society development [1]. Thus, the objectives are not limited to the economic impact of logistics activities and operations, but also to the broader impact on the environment and society, including environmental pollution measurement, waste collection, and customer satisfaction analysis [2]. Green logistics serves as a strategy for businesses to achieve sustainability, ensuring long-term profitability in logistics operations. The implementation of green logistics (GL) involves a variety of activities and operations. Zhang et al. [3] introduces a framework based on GL, which is structured into three main categories: green performance measurement, green operations execution, and green strategy formulation. Network design is one of the sub-dimensions of green operations. Indeed, upon reviewing the literature, the number of VRP publications accounts for nearly 50 % of the category of network design, emphasizing its crucial role in both research and practical implementations [3].

Numerous VRP versions have been developed since the introduction of VRP, considering various conditions, application areas, and practical considerations. Green vehicle routing problem (GVRP), as identified by [4], represent VRP variations specifically addressing sustainable transportation issues. The goal of GVRP is to balance environmental and economic costs by developing effective routes to address environmental and financial concerns. Futhermore, the triple bottom line of sustainability facilitates a comprehensive understanding of GVRP application by encompassing economic aspects through cost reductions, environmental considerations involving GHG emissions and fuel consumption, and social factors such as working conditions and opportunities [5]. The

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minimization of emissions may sometimes conflict with economic objectives like reducing overall costs, total travel distances, or travel time. Consequently, GVRP involves multiple objectives that need simultaneous fulfillment, prompting researchers to seek viable and high-quality solutions for specific optimization challenges.

Several exact and approximation approaches have been developed within the perspective of operations research to tackle various variants of the green vehicle routing problem. According to the findings by [6], approximately 41 % of the researchers propose metaheuristic approaches, with only 14 % incorporating hybrid algorithms. Hybridization attempts to utilize the benefits of each method to create a hybrid algorithm while minimizing any significant drawbacks [7]. Typically, hybridization leads to improvements in either computational speed or accuracy. According to Erdelic and Carić [8], a large number of hybrid algorithms have been proposed for addressing various VRPs. As a result, the exploration of hybrid approaches in GVRP merits further investigation.

In the past three years, Moghdani et al. [6], Macrina et al. [9], Marrekchi et al. [10], Asghari and Mirzapour Al-e-hashem [11] and Gil et al. [12] have classified previous studies into single-objective and multi-objective GVRP. They have also reviewed the solution approaches and algorithms employed by each researcher. However, to the best of our knowledge, only [6] and [12] have thoroughly and explicitly differentiated the solution approaches and algorithms used in single-objective and multi-objective GVRP. Therefore, this review study aims to strengthen two previous studies by conducting an in-depth analysis of the differences in solution approaches employed to address the GVRP problems, both single-objective and multi-objective. Additionally, it will provide a detailed review of the algorithms used within each solution approach.

Optimization software and programming languages used for solving mathematical models and proposed algorithms have only been reviewed by [5]. However, [5] focuses specifically on the solution approach and algorithm for addressing multi-objective GVRP. Moreover, no previous study has systematically reviewed the solution approaches, particularly published papers addressing GVRP after 2021. Hence, this study aims to identify and address the gaps in existing review studies. Therefore, the main contributions of this study are outlined as follows:

1. An up-to-date and comprehensive review of the GVRP and its variants, with a focus on analyzing the problem characteristics, mathematical models, solution approaches, solution algorithms, optimization software, and programming languages proposed by researchers.
2. A new presentation of the relationship between the mathematical model formulated and the solution approach proposed by GVRP researchers.
3. A comprehensive study of solution approaches and algorithms for addressing both single-objective GVRP and multi-objective GVRP. The discussion is differentiated for each type of optimization, consisting of single-objective and multi-objective. Additionally, to enhance the study, we specifically highlight type of hybrid approach and the name of algorithms incorporated in hybrid approach.
4. A novel analysis of solution approaches that haven't been extensively covered in previous research. This study introduces insights into hyper-heuristics, hybrid heuristics-heuristics, and hybrid optimization-simulation approaches.

The remainder of the paper is organized as follows. Section 2, reviews of previous review studies. In Section 3, describes the review methodology. Section 4 explains the problem characteristics of GVRP. Section 5 provides an explanation of the solution methodology used in each GVRP research. An in-depth analysis of the proposed algorithms for the GVRP is given in Section 6. Finally, Section 7 presents the conclusions of this work and future research directions.

2. A review of previous review studies

The increasing number of research and applications in GVRP has been highlighted in various review studies. According to Lin et al. [4], GVRP can be divided into three broad categories: green-VRP (G-VRP), pollution routing problem (PRP), and VRP in reverse logistics (VRP-RL). Table 1 indicates that the majority of these studies have primarily focused on specific categories of GVRP. Reviews conducted by [11, 13-18] have focused on G-VRP, while [6,9,10,19-21] have focused on both G-VRP and PRP. It is important to mention that only [5,12] and [24] have conducted comprehensive studies encompassing all three categories. This review study will enrich the findings of these three previous studies by incorporating GVRP researches published in 2022 and 2023.

Regarding the problem domain, discussions have primarily focused on the types of fleets and fuels. From Table 1, it can be observed that 10 out of the 18 previous review studies have extensively discussed the types of fleets utilized in GVRP research. Furthermore, the review studies have thoroughly addressed the utilization of both fossil and non-fossil fuels in GVRP research. For those interested in an in-depth review of GVRP classification based on fuel types, we recommend the review studies conducted by [9] and [11]. Macrina et al. [9] introduced a classification framework based on various G-VRP variants found in the scientific literature, identifying two primary categories: (1) G-VRP with conventional vehicles and (2) G-VRP with alternative fuel vehicles (AFVs). The first category comprises four subcategories: time-dependent PRP, multi-objective PRP, heterogeneous fleet PRP, and location PRP, while the second category includes six subcategories: G-VRP with AFVs, G-VRP with EVs, G-VRP with HEVs, mixed-fleet G-VRP, G-VRP with EVs and locations, and G-VRP with EVs and nonlinear charging function. Additionally, Asghari and Mirzapour Al-e-hashem [11] presented an overview of three significant and relevant streams of green-VRPs: internal combustion engine vehicles (ICEVs), alternative fuel vehicles (AFVs), and hybrid electric vehicles (HEVs).

In optimization problems, the formulation of mathematical models is a crucial preliminary step. Previous studies analyzing the mathematical models formulated by GVRP researchers have been conducted only by [5,9,10,19,21]. Therefore, this review will also discuss the mathematical models used by GVRP researchers. As this review study addresses solution approaches for both single-objective and multi-objective GVRP, it is important to discuss the number of objective functions. As shown in Table 1, the majority of review studies have also categorized GVRP research based on the number of objective functions. Furthermore, we also review optimization software and programming languages commonly used to solve the green vehicle routing problem (GVRP). Programming languages commonly utilized by researchers likely have various libraries and existing tools that can be utilized to implement GVRP algorithms. Thus, the findings of this study review can provide insights for future GVRP research.

Among the 18 previous review studies, 16 have discussed the solution approaches proposed by GVRP researchers. Regarding the types of solution approaches, the majority of previous review studies have classified the proposed solution approaches into exact, heuristic, metaheuristic, and hybrid. Moghdani et al. [6] categorized the solution approaches into three main groups, consisting of exact (including exact and exact solver), approximation (encompassing metaheuristic, heuristic, and hybrid algorithms), and others (including software applications), to provide a comprehensive overview of previous studies on GVRP and its variants. Regarding hybrid approaches, they encompass exact-metaheuristics, exact-heuristics, metaheuristics-metaheuristics, and metaheuristics-heuristics. Moreover, Gil et al. [12] explored more deeply the use of heuristics and hybrid approaches to solve GVRP considering emissions. For hybrid metaheuristics, the classifications include hybrid metaheuristics-heuristics, hybrid metaheuristics-meta heuristics, hybrid metaheuristics with mathematical programming, and hybrid metaheuristics for multi-objective optimization. We also

Table 1
Summary of previous review studies.

Publication		Number of papers reviewed	Data coverage	The main featuress that have been reviewed										
Author	Year			GVRP category	Fleet type	Vehicle fuel	MM	Obj description	Obj function	Solution approach type	Combinati-on type in hybrid	Optimiza-tion software	Programming language	Bench-mark instances
Demir et al. [19]	2014	58	2006–2013	G-VRP, PRP	Ho, He	GDV	v	v	Single, Multi	E, H, M, SA	–	–	–	–
Park and Chae) [13]	2014	40	2000–2014	G-VRP	–	GDV	–	–	–	E, H, M	–	–	–	–
Lin et al. [4]	2014	284	1959–2013	G-VRP, PRP, VRP-RL	–	GDV	–	–	–	–	–	–	–	–
Marrekchi et al. [21]	2019	28	2014–2019	G-VRP, PRP	–	GDV	v	–	–	E, H, M	–	–	–	–
Asghari and Mirzapour Al-e-hashem [14]	2020	236	1976–2019	G-VRP	Ho, He	GDV, EV, AFV, HEV, MFV	–	v	Single, Multi	E, H, M, Hb	–	–	–	–
Ferreira et al. [5]	2020	76	2012–2018	G-VRP, PRP, VRP-RL	Ho, He	GDV	v	v	Multi	E, H, M, Hb	–	v	v	–
Ghorbani et al. [15]	2020	125	2011–2020	G-VRP	–	EV, AFV, HEV, MFV	–	v	–	E, H, M, Hb, ES	–	–	–	v
Macrina et al. [9]	2020	75	2011–2019	G-VRP, PRP	Ho, He	GDV, EV, AFV, HEV, MFV	v	–	Single, Multi	E, H, M, Hb	–	–	–	–
Asghari and Mirzapour Al-e-hashem [11]	2021	313	2000–2020	G-VRP	Ho, He	GDV, AFV, EV, HEV	–	v	Single, Multi	E, H, M, Hb	–	–	–	–
Kucukoglu et al. [17]	2021	136	2011–2021	G-VRP	Ho, He	EV	–	v	Single, Multi	E, H, M, Hb	–	–	–	v
Marrekchi et al. [10]	2021	86	2014–2019	G-VRP, PRP	Ho, He	GDV	v	v	Single, Multi	E, H, M, Mh	–	–	–	v
Martins et al. [22]	2021	105	2012–2021	G-VRP	Ho, He	EV	–	–	–	E, H, M, Hb, S, ML	–	–	–	–
Moghdani et al. [6]	2021	309	2006–2019	G-VRP, PRP	–	GDV, EV	–	v	Single, Multi	E, H, M, Hb, SA, ES	v	–	–	–
Qin et al. [16]	2021	49	2012–2020	G-VRP	Ho, He	EV	–	–	–	E, H, M, Hb	–	–	–	–
Gil et al. [12]	2022	89	2017–2021	G-VRP, PRP, VRP-RL	Ho, He	GDV, EV, AFV	–	–	Single, Multi	H, M, Hb	v	–	–	v
Matijevic [20]	2022	62	2013–2021	G-VRP, PRP	–	GDV, EV, AFV, HEV, MFV	–	v	Single, Multi	M, Hb, ES	–	–	–	–
Sabet and Farooq [18]	2022	450	1990 - 2020	G-VRP	–	GDV, EV, AFV, HEV, MFV	–	–	–	E, H, M, Hb, SA, ES	–	–	–	–
Li et al. [23]	2023	1230	2000–2021	G-VRP, PRP, VRP-RL	–	GDV, EV, AFV	–	–	–	–	–	–	–	–
Our review	2024	458	2016–2023	G-VRP, PRP, VRP-RL	Ho, He	GDV, EV, AFV, HEV, MFV	v	v	Single, Multi	E, H, M, Hb, ES	v	v	v	–

GDV: Gasoline or diesel vehicle, AFV: Alternative fuel vehicle, EV: Electric vehicle, HEV: Hybrid vehicle, MFV: Mixed fleet vehicle, Ho: Homogeneous, He: Heterogeneous, MM: Mathematical model, E: Exact, H: Heuristic, M: Metaheuristics, Hb: Hybrid, SA: Software application, S: Simulation, ML: Machine Learning, ES:Exact solver, v: This feature is discussed in the paper, -: This feature is not discussed in the paper.

provide detailed discussions regarding the hybrid approaches employed by previous GVRP studies, introducing new types of hybrid approaches: hybrid heuristics-heuristics and hybrid simulation-optimization.

This review study specifically focuses on detailing the algorithms incorporated in each solution approach, the hybrid approach types, and the algorithms utilized within hybrid approaches. Additionally, the findings from [12] serve as a valuable reference for better understanding the leading strategies and methodologies employed in various solution approaches. These encompass initial solution procedures, the structure of neighborhoods in a heuristic scheme, and the operators employed in evolutionary approaches. Moreover, there has been a growing trend using hyper-heuristics as a solution approach for vehicle routing problem [24]. Futhermore, this review study introduces a new analysis for solution approaches in GVRP, known as hyper-heuristics. Additionally, certain GVRP studies that do not address real-world problems utilize datasets from benchmark instances. Given that GVRP is part of the VRP family, there is a set of instances originally developed for VRP and subsequently adopted in GVRP. Among the 18 review studies, only those conducted by [10,12,15,17] addressed benchmark instances.

3. Review methodology

To conduct this comprehensive review, we systematically searched articles related to the green vehicle routing problem in the Web of Science (WoS) Core Collection database. We utilized specific syntax keywords including 'green vehicle routing', 'pollution vehicle routing', 'vehicle routing in reverse logistics', 'electric vehicle routing', 'alternative fuel vehicle routing', 'carbon emission vehicle routing', or 'waste collection vehicle routing'. The search spanned from January 2016 to December 2023. The literature search conducted through the WoS database was complemented by further searches using the Google Scholar search engine. Following the collection of articles and a review of their titles, a total of 150 duplicate articles were identified and subsequently removed from both searches. Subsequently, a screening process based on title and abstract was conducted to exclude studies not related to the green vehicle routing problem. A total of 861 articles were excluded based on this analysis, resulting in 520 records. In the final step, we thoroughly examined the full texts of the 520 papers, and despite the removal of some, a final selection of 468 papers was considered for this comprehensive review. Fig. 1 illustrates the employed search strategy and presents the records acquired at each step of the process.

4. Problem characteristics

This section provides a detailed overview of the classification of papers based on the characteristics of the problem. This classification encompasses several key aspects, including the GVRP category, fleet type, vehicle fuel type, GVRP variants, number of objective functions, and objective function type.

4.1. Category of GVRP

Sbihi and Eglese [25] were among the pioneers in incorporating environmental objectives into routing problems. Kara et al. [26] introduced the initial model focusing on reducing fuel consumption by minimizing a weighted distance function, establishing the first version of the energy minimization vehicle routing problem (EMVRP). Until 2011, only a limited number of studies implicitly addressed environmental concerns. The pollution routing problem (PRP), proposed by [27], emerged as a new variant of the vehicle routing problem with time windows (VRPTW). PRP aims to optimize vehicle speed on each arc to minimize fuel consumption, emissions, and driver costs. Notably, the PRP allows for variations in load and speed between different arcs while keeping other parameters constant. To the best of our knowledge, Erdogan and Miller-Hooks [28] were the first to introduce the green vehicle routing problem (G-VRP), a variation of the VRP that accounts for the additional challenges associated with operating a fleet of alternative fuel vehicles (AFVs). Since its introduction, G-VRP has been applied to a wide range of vehicles, including diesel-powered vehicles, electric vehicles, and hybrid vehicles.

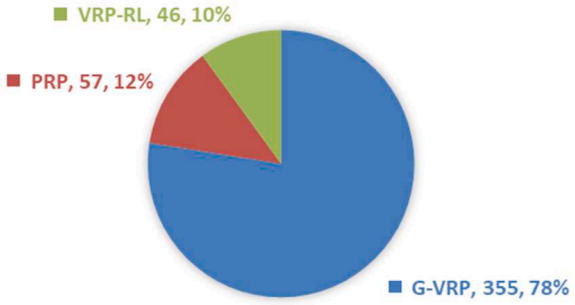


Fig. 2. The number of papers in each GVRP category.

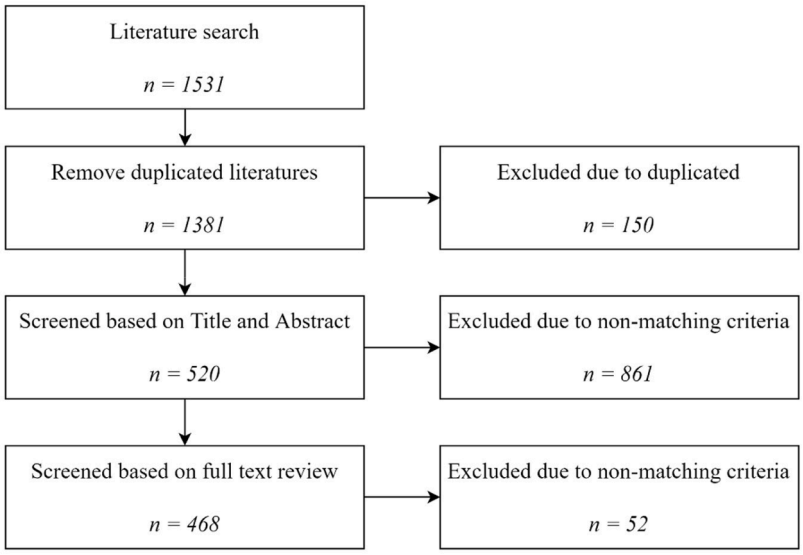


Fig. 1. The steps of searching papers for review.

Fig. 2 illustrates the classification of the 458 papers that have been reviewed, referring to the categories proposed by [2]. The G-VRP category is a field of study that concentrates on two primary aspects (1) optimizing fuel consumption through considerations such as transportation distance, vehicle speed, and load weight, and (2) exploring the utilization of alternative energy sources, including electricity, biodiesel, compressed natural gas (CNG), and hydrogen. Given the expansive scope of this category, it is unsurprising that 355 papers are encompassed within it. Additionally, 57 papers fall under the pollution routing problem (PRP) category, while 46 papers are categorized under the vehicle routing problem in reverse logistics (VRP-RL) category.

As depicted in Fig. 3, the GVRP has gained growing attention within the scientific community over the years. The regression equation representing the correlation between the number of GVRP papers (Y) and the respective years (X) is a expressed as 5.381X + 33.036. Despite minor annual fluctuations in the number of GVRP studies, the positive slope of the regression suggests an increasing trend of studies on GVRP. Based on the paper published in 2016–2023, we find that there are a lot of interesting publications on GVRP proposing new tools of optimization and new formulations.

4.2. Type of fleet

Fig. 4 indicates that although a significant portion, accounting for 67.47 %, comprises a homogeneous fleet of vehicles, there is also considerable attention towards heterogeneous fleet, amounting to 32.31 %. It is common for companies to employ vehicles from diverse brands featuring different engines, potentially resulting in variations in emissions levels and fuel types used. Moutaoukil et al. [29] found that the use of a heterogeneous fleet gives the best environmental results and presents a favorable balance between economic and social costs compared to utilizing a homogeneous fleet. Sun et al. [30] developed an exact algorithm to solve the problem of heterogeneous green pickup and delivery (GPDP). They found that employing a heterogeneous fleet can reduce carbon emissions by around 19 %. Rezaei et al. [31] proposed a bi-objective MILP model that minimizes total transportation costs and CO_2 emissions. Their comprehensive sensitivity analysis revealed that employing a heterogeneous fleet not only reduces transportation costs but also mitigates pollution emissions. Notably, in Fig. 4, only one paper does not provide information about the type of vehicle. Ebrahimi [32] developed a closed-loop supply chain network for the tire industry considering sustainability aspects and quantity discounts. While various potential routes are assessed for shipping products from manufacturers to distribution centers (DCs), the study lacks information on the type and capacity of the vehicles used.

There are two categories of VRP considering a heterogeneous fleet of vehicles. First, the fleet size and mix vehicle routing problem (FSMVRP), as proposed by [33], which deals with an unlimited heterogeneous fleet. Second, the heterogeneous vehicle routing problem (HVRP) introduced by [34], which works with a predetermined fleet. When compared, the FSMVRP stands out as the more prevalent category due to its simplicity and practicality. Alinaghian and Zamani [35] presented a FSMVRP that minimizes a total cost comprising of fuel consumption, pollutant emissions, fixed vehicle costs, and driver wages. After formulating a Mixed-Integer Linear Programming (MILP) model, they introduced three heuristics to address the problem. On the other hand, research on the Heterogeneous GVRP, conducted by [30,31,36-52]. Ghannadpour and Zarrabi [46] introduced a novel model and solution for the multi-objective heterogeneous vehicle routing and scheduling problem, taking fuel consumption into account. Li et al. [49] studied an emission-based heterogeneous fixed fleet vehicle routing problem (E-HFFVRP) considering fuel consumption and carbon emissions. This problem involves routing a fixed number of vehicles with various capacities to serve a set of customers.

4.3. Type of vehicle fuel

As depicted in Fig. 5, there are 311 papers classified under the gasoline or diesel vehicle (GDV) category. This signifies that almost 68 % of researchers remain engaged in investigating GVRP utilizing fossil fuels or conventional vehicles. In contrast, the number of GVRP studies involving non-fossil fuel vehicles from 2016 to 2023 stands at 147 papers. Among these, 80 papers focused on electric fuel, 23 studies explored alternative energy sources apart from electricity, 8 studies involved hybrid vehicles, and 30 papers introduced mixed fleet vehicles.

Ashtineh and Pishvaee [53] explored and presented insights into the economic and environmental implications associated with various alternative fuel vehicles (AFVs). In their study, each AFV within a route originates from a shared depot where the fleet is stationed, serves a specific subset of customers, and subsequently returns to the depot, possibly making fueling stops at alternative fuel stations (AFSs). However, AFVs encounter certain limitations such as restricted storage capacity due to the low-capacity batteries and limited accessibility to AFSs. Consequently, addressing the routing problem of AFVs is complicated by the need to strategize vehicle recharging or refueling and design an appropriate charging infrastructure [54].

To tackle the Green Vehicle Routing Problem incorporating AFVs, Koç and Karaoglan [55] proposed a solution approach based on simulated annealing heuristics. In 22 out of 40 test instances, their findings revealed the identification of optimal solutions. Affi et al. [56] addressed the G-VRP, focusing on determining the optimal times and locations for refueling or recharging the AFV fleet to minimize total energy costs. Additionally, Bruglieri et al. [57] developed a novel path-based MILP models, eliminating the need for path cloning, for both private and public AFS scenarios. They further enhanced their approach to encompass both scenarios through a more efficient cutting planes approach. Many studies in this category have been carried out by [51,58-63].

The environmentally friendly aspects, such as low noise pollution, zero GHG emissions, and high energy efficiency, are the primary reason for the growing popularity of electric vehicles (EVs) [64]. Beyond the

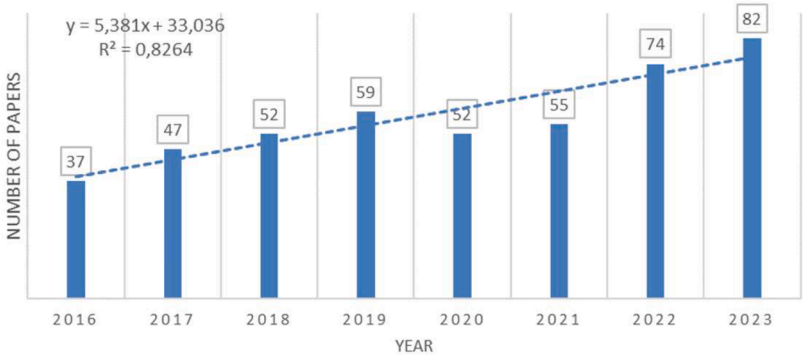


Fig. 3. The number of GVRP paper publications each year.

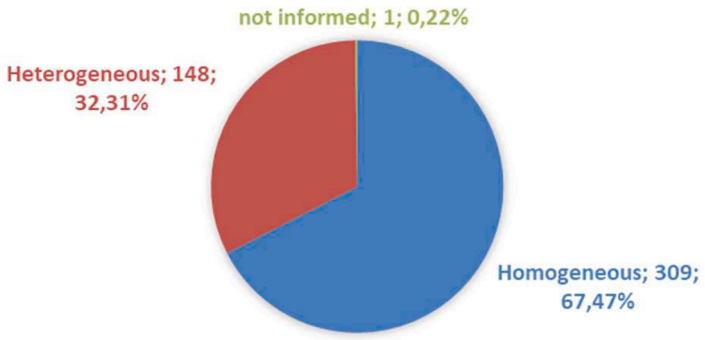


Fig. 4. Type of fleet.

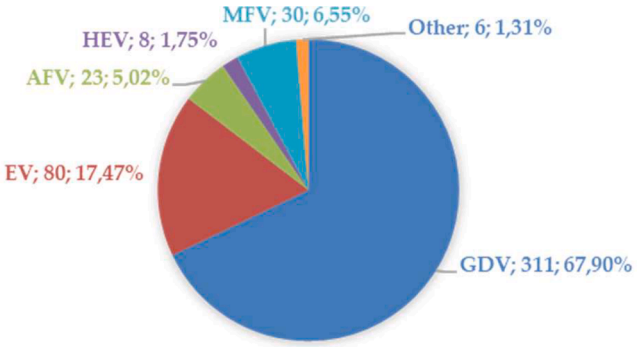


Fig. 5. Type of vehicle fuel.

environmental advantages, EVs are considered more cost-effective compared to conventional vehicles. When comparing the costs of gasoline or diesel fuel consumed by internal combustion engine vehicles (ICEVs), EV costs are reduced by 85 to 90 % for the same distance traveled [11]. However, challenges such as battery duration and pricing, driving distance, and the replacement of conventional vehicles remain significant barriers for the widespread adoption of EV fleets. Desaulniers et al. [64] introduced four variants of EVRP with time windows, considering factors like the number of recharges (single and multiple) and full or partial recharges. They employed an exact branch-price-and-cut algorithm to derive optimal solutions. Zhao and Yu [65] presented an environmentally sustainable model of EVRP, encouraging logistics companies to incorporate electric vehicles into their operations.

Hybrid vehicles have the flexibility to derive the energy needed for their movement from two sources: electric fuel and either gasoline or diesel fuel. These vehicles have the capability to select and switch between these fuel options at any given moment [66]. Despite the general preference for hybrid vehicles over other types, literature reveals that studies concentrating specifically on hybrid electric vehicles (HEVs) are the least represented among various vehicle types, as indicated in Fig. 5. However, the expanding market for HEVs supports the practicality of hybrid electric vehicle routing problem (HEVRP), typically oriented towards minimizing the total cost associated with employing hybrid vehicles. Significant contributions to the HEVRP field are found in the studies conducted by [66-70]. Yu et al. [67] introduced the HVRP, focusing on vehicles utilizing a hybrid power source, specifically the plug-in hybrid electric vehicle (PHEV).

A mixed fleet incorporates various vehicle types for distribution simultaneously, including EVs, AFVs, and GDVs. The utilization of mixed fleets is regarded more applicable to real-world conditions due to differences in speed, price, equipment, technology, and capacity [2]. With the growing prevalence of EVs in distribution systems, optimizing routes becomes crucial for cost-effective interactions among different vehicle types [15]. Prominent research exploring the integration of mixed fleets includes works by [71-81]. Islam and Gajpal [73] demonstrated that optimizing the composition of conventional and green vehicles can reduce CO_2 emissions by 6.9 % compared to fleets consisting solely of conventional vehicles. Macrina et al. [79] proposed an iterative local search heuristic for optimizing the routing of a mixed vehicle fleet, comprising electric and conventional vehicles. Islam et al. [81] presented the mixed fleet-based green clustered logistics problem (MFGCLP) under a CO_2 emission cap to deal with the sustainable development efforts of the transportation industry. The mixed fleet in this scenario consists of hydrogen vehicles and conventional vehicles. Additionally, Macrina et al. [82] investigated the management of EVs and conventional vehicles to reduce costs associated with routing and recharging operations.

4.4. GVRP variant

Numerous variants of the classic VRP have been implemented in the context of the green vehicle routing problem (GVRP). As indicated in Fig. 6, time windows are the most common variant, followed by location routing, pickup and delivery, time-dependent, and multi-echelon network. In 2008, Kara et al. [83] introduced the cumulative vehicle routing problem (CumVRP) as a pertinent routing variant designed to incorporate environmental considerations. The primary objective is to minimize the cumulative cost function, facilitating the computation of emissions based on the distance covered by a vehicle carrying a specific load and the load realization throughout the route [84]. In this context, CumVRP falls under the category of problems in the green routing domain [85], as it accounts for the gradual reduction in vehicle load during the delivery process, resulting in lower CO_2 emissions as the

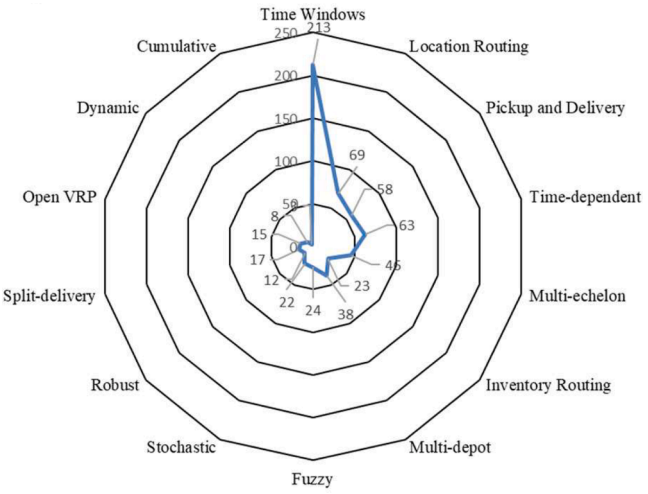


Fig. 6. GVRP variants.

vehicle traverses its route. Therefore, CumVRP is added to the discussion on GVRP variants considering its role in GVRP.

To address the uncertainties inherent in the real world, researchers employ various approaches including fuzzy, stochastic, and robust optimization. As shown in Fig. 6, it is apparent that 24 researchers have introduced fuzzy GVRP, while 22 researchers have presented stochastic GVRP. In subSections 4.4.1 to 4.4.4, a comprehensive study of several GVRP variants will be provided.

4.4.1. GVRP with time windows

As shown in Fig. 6, time windows are considered in 213 papers, accounting for 46.51 % of the total. Rezaei et al. [31] examined the impacts of tightening and widening time windows on solutions. Costs and CO_2 emissions increase as time window durations are tightened, and conversely, decrease as they are widened. Excessively tight time windows may render the problem unsolvable, as vehicles may be unable to meet all demands within the specified time constraints. Xiao et al. [86] applied an iterative neighborhood search heuristic and proposed a new operator, called time neighbor selection, to choose customer nodes that have neighbored time windows. Customers who are closer in terms of time window (although may be far apart in terms of distance) are more likely to be serviced sequentially. Wang et al. [87] studied a bi-objective vehicle-routing problem with soft time windows and multiple depots, with the aim of minimizing total energy consumption and customer dissatisfaction at the same time. Furthermore, Xu et al. [88] presented a model addressing the capacitated green vehicle routing problem incorporating time-varying vehicle speed and soft time windows. The objective of the model is to maximize customer satisfaction while minimizing overall fuel consumption, equivalent to carbon emissions.

4.4.2. Green location-routing problem

In designing an effective distribution system all decisions concerning transportation, routing, and the positioning of intermediary facilities such as warehouses or urban distribution centers must be optimized [89]. The integration of the facility location problem (FLP) and the vehicle routing problem (VRP) results in the location-routing problem (LRP). The green location-routing problem (GLRP) expands upon the traditional location-routing problem, focusing on minimizing fuel costs and reducing CO_2 emissions during both manufacturing and distribution processes. As depicted in Fig. 6, 69 out of 458 papers (15.07 %) have proposed model and solution approaches for GLRP. Vakili et al. [90] formulated the green open location-routing problem using stochastic programming and robust optimization, where travel time and customer demands include pick-up and delivery demands assumed to be probabilistic. Ziaei and Jabbarzadeh [91] presented a multi-objective model for determining the optimal locations of transfer points and routing in a multi-modal network handling hazardous materials. This model takes into consideration various uncertainties, aiming to minimize carbon emissions, transportation risk, and total costs simultaneously. Tirkolaee et al. [92] formulated the sustainable multi-trip location-routing problem with time windows (MTLRP-TW) for medical waste management in the COVID-19 pandemic. The objectives are to simultaneously minimize the total travel time, manage violations of time windows and service priorities, and reduce the overall infection or environmental risk to the population residing near disposal sites.

4.4.3. Green time-dependent vehicle routing problem

The time-dependent vehicle routing problem (TDVRP) is a variant of the traditional VRP, considering traffic conditions and assuming that the speed and travel time on each route segment depend on the specific time of traversal. In real-world routing problems, travel times on individual arcs are often stochastic due to unpredictable factors such as bad weather, car accidents, and special events causing traffic congestion [93]. Time-dependent plays a significant role in contributing to emissions in road transportation, as evidenced by the inclusion of 63 papers (13.76 %). Hooshmand and MirHassani [94] proposed a novel variant of TDVRP considering alternative green-fuel-powered vehicles with the objective of minimizing CO_2 emissions. Implementing a time-dependent formulation led to an average decrease of 11.71 % in comparison to the time-independent approach. Soysal and Çimen [95] developed a model for the green time-dependent vehicle routing problem to explicitly incorporate time-dependent speeds and fuel consumption. Numerical experiments revealed that traffic congestion negatively impacts both total emissions and total routing costs. Other previous studies that considered time-dependent have been conducted by [96-103].

4.4.4. Green pickup and delivery problem

The traditional VRP assumes that customers only have a demand for delivery. However, in real-world scenarios, customers may have both pickup and delivery demands, often requiring fulfillment simultaneously. Recognizing this demand structure, a variant of VRP known as the VRP with pickup and delivery (VRPPD) was introduced. We focused on a specific variation of VRPPD, considering environmental factors, known as the green pickup and delivery problem (Green-PDP). Kunnapapdeelert and Klinsrisuk [104] conducted a comparison between pickup and delivery and the green pickup and delivery problem using a differential evolution algorithm. They demonstrated that planning routes with an emphasis on minimizing GHG emissions results in a more environmentally friendly design compared to routes planned solely based on minimizing total cost.

Several studies by [105-111] have been conducted to tackle Green-PDP, which objectives to reduce fuel consumption and CO_2 emissions. Abad et al. [105] introduced a bi-objective programming model for PRP with pickup and delivery, which integrates and consolidates shipments in a cross-docking system. Majidi et al. [111] formulated a mixed-integer nonlinear programming model for PRP, incorporating simultaneous pickup and delivery and utilizing an ALNS heuristic for its solution. Furthermore, various researchers have explored extensions of Green-PDP, some of which integrate with facility location problems. Research on green LRP with pickups and deliveries has also been conducted by [90,112-116]. Leng et al. [113] proposed a comprehensive optimization model based on cold-chain principles to minimize total logistics costs and waiting time for vehicles and clients. Wang et al. [114] investigated into the green logistics location-routing problem (GLLRP), with a particular focus on eco-packages.

4.5. Objective function

4.5.1. Number of objective functions

Fig. 7 illustrates an increasing trend in the number of studies concerning both single-objective GVRP and multi-objective GVRP from 2016 to 2023

Building on the insights from Fig. 8, it becomes apparent that single-objective optimization emerges as the predominant approach in addressing the green vehicle routing problem. A substantial majority, 303 out of 458 papers (66.16 %), belong to this category. Additionally, among the 156 papers addressing multi-objective optimization, 96 focus on bi-objectives, while 59 papers incorporate more than two objectives. Fig. 8 also illustrates the distribution of the number of objectives in each GVRP category. For example, in the G-VRP category, 237 papers focus on a single objective function. Furthermore, 75 papers investigate bi-objective GVRP, and 36 papers propose more than two objectives.

4.5.2. Objective function type

The majority of GVRP models typically focus on the environmental and economic aspects of routing problems, often neglecting social aspects in existing literature [11]. These challenges have led to the emergence of a new concept known as the sustainable routing problem, aiming to strike a balance across the three pillars of sustainability: economic, environmental, and social dimensions. Based on the review of 458 papers, we classified the objective functions into 28 categories, as outlined in Table 2. Objective function codes 1–14 represent the

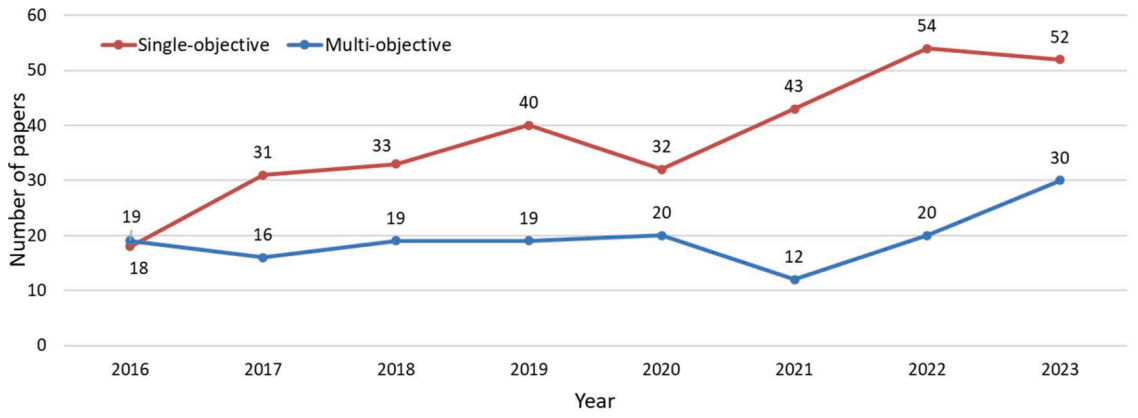


Fig. 7. The distribution of single-objective GVRP and multi-objective GVRP each year.

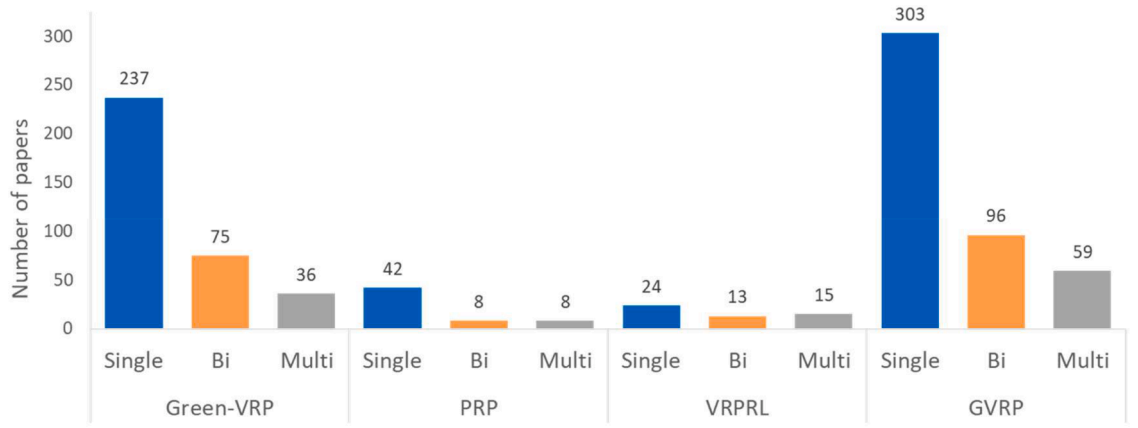


Fig. 8. Number of objectives in each category of green vehicle routing problem.

Table 2
Objective function notation.

Code	Objective function	Sustainability dimension
1	Charging cost	Economic
2	Driver wage and vehicle cost	Economic
3	Energy used or cost	Economic
4	Fuel consumption	Economic
5	Number of vehicles	Economic
6	Operating cost	Economic
7	Profit or revenue	Economic
8	Recharging time	Economic
9	Total cost	Economic
10	Total cost including battery swapping cost	Economic
11	Total distance or travel cost	Economic
12	Total travelling time	Economic
13	Travel cost and charging cost	Economic
14	Travel cost and refueling cost	Economic
15	Driver wage and emission cost	Economic & Environmental
16	Operating cost including emission cost	Economic & Environmental
17	Total cost including emission cost	Economic & Environmental
18	Travel cost and emission cost	Economic & Environmental
19	Total CO_2 emission or emission cost	Environmental
20	Customer satisfaction	Social
21	Lost sales	Social
22	Reliability or responsiveness	Social
23	Social impact	Social
24	Tardiness or earliness	Social
25	Transportation risk	Social
26	Waiting time or cost	Social
27	Workload balancing	Social
28	Others	

economic dimension, while objective function code 19 represents the environmental dimension. Through the review of the most relevant papers, the social concept has emerged, addressing issues such as human resources, safety concerns, and customer satisfaction. These issues are accommodated by objective function codes 20–27. Furthermore, the review results indicate that some researchers combine objective functions that encompass both economic and environmental dimensions in their single-objective GVRP model. They propose an objective function that accounts for both dimensions, represented by objective function codes 15–19.

Fig. 9 presents the distribution of each objective function proposed by GVRP studies. Following a review of 458 papers, the top five objective functions and their respective percentages are as follows: total CO_2 emission or emission cost (17.8 %), total cost (14.2 %), total cost including emission cost (12.4 %), total distance or travel cost (11.3 %), and fuel consumption (7.1 %). These five objective functions account for 62.8 % of the research, indicating a predominant focus on the environmental and economic dimensions. In contrast, the cumulative percentage of objective functions in the social dimension, represented by codes 22–27, amounts to 10.9 %. This highlights that, overall, researchers tend to prioritize the social dimension as the least addressed objective function.

Fig. 10 indicates that various economic and environmental objective functions have been explored in the context of single-objective GVRP. On the contrary, objective functions associated with the social dimension are rarely considered in single-objective GVRP. From Fig. 10, it can be seen that minimizing lost sales is proposed by 3 researchers. The lack of consideration for social dimension aspects in single-objective GVRP may be attributed to the predominant focus of GVRP on economic and

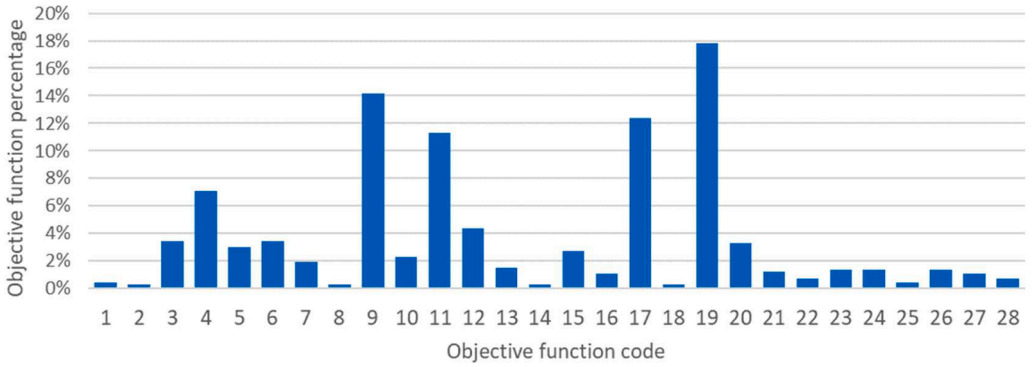


Fig. 9. Objective functions in GVRP models.

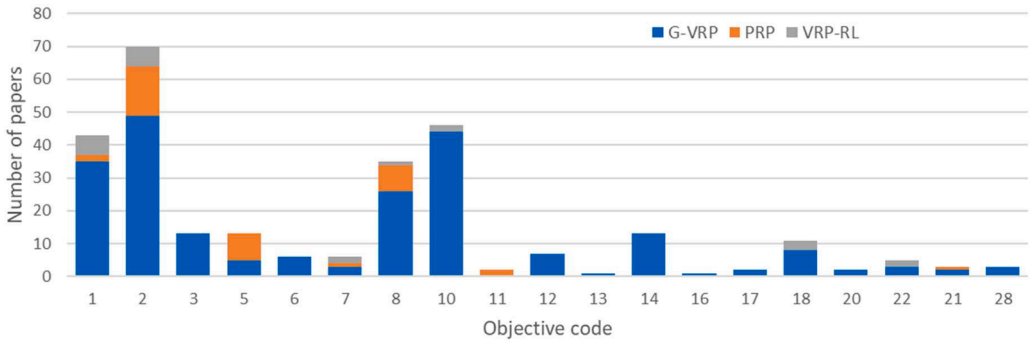


Fig. 10. Objective functions in single-objective GVRP.

environmental dimensions. In single-objective G-VRP, total cost including emission cost is an important objective function besides total distance or travel cost. Fig. 10 further illustrates the distribution of objective functions within each GVRP category. In the G-VRP category, researchers most commonly propose objective functions consisting of total cost including emission cost and total distance/travel cost. For PRP category, the predominant objective functions are total cost including emission cost and total CO_2 emission/emission cost. Lastly, for VRP-RL, the two most commonly proposed objective functions are total cost and total cost including emission cost.

Fig. 11 presents the distribution of objective functions in multi-objective GVRP. The most commonly proposed objective function is the total emission cost, which represents the environmental dimension. Furthermore, the subsequent highly regarded objective function in multi-objective GVRP is the total cost, which represents the economic dimension. This indicates that the objective functions of minimizing total cost and emission cost, which are typically combined as a single objective function in single-objective GVRP, become standalone objective functions in multi-objective GVRP. In multi-objective GVRP, researchers suggest a broader spectrum of objective functions, covering codes 1 to 28. As depicted in Fig. 11, objective functions with codes 20 (customer satisfaction), which represent the social dimension, are commonly proposed by GVRP researchers. Additionally, frequently considered objective functions include social impact, tardiness/earliness, and waiting time/cost.

5. Solution methodologies

5.1. Mathematical model formulation

Optimization problems are a special type of problem in which the goal is to maximize or minimize a single-objective or multi-objective function of a set of decision variables while ensuring compliance to specified constraints. The decision variables in the model may be real or

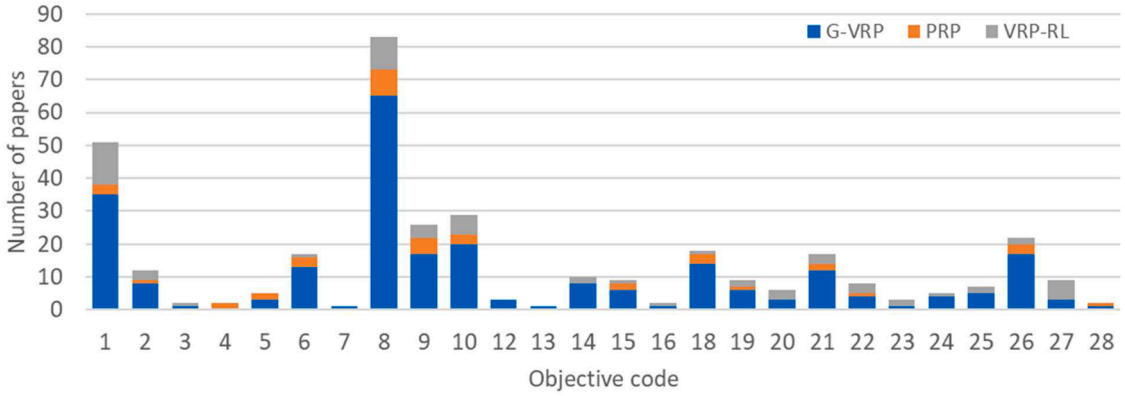


Fig. 11. Objective functions in multi-objective GVRP.

integer (pure integer or binary integer) or a mix of both. Depending on the nature of the problem, the objective or constraint functions may be either linear or nonlinear. If all the functions in a given model are linear, it is referred to as a linear programming model. Whereas a model is called a nonlinear model if one or more of its functions are nonlinear [117]. The literature has identified various classifications of mathematical models based on variable and function categorization, which include integer linear programming (ILP), mixed integer linear programming (MILP), and mixed integer nonlinear programming (MINLP). The predominant form of mathematical models employed for addressing the GVRP issue is MILP. However, there are researchers who propose the use of MINLP, as evidenced by several studies conducted [69,89,118-120].

Based on the findings from reviewing the MM formulation, the MILP and MINLP models developed by researchers have been deterministic thus far, with parameters known with certainty. However, real-world situations typically involve unknown or uncertain parameters, such as demand, travel time, vehicle speed, and distance. Optimizing a mathematical model with uncertain parameters is inherently complex, requiring the utilization of a specific method. To address this complexity, various methods have been employed in optimization modeling, including stochastic programming, fuzzy programming, basic recourse-based stochastic, chance-constraint programming, and robust optimization. Araghi et al. [121] developed a new mathematical model, the green stochastic open location-routing problem (GSOLRP), which addresses uncertain parameters in the objective function. They proposed a robust worst-case optimization approach for the GSOLRP with demand from a probabilistic customer location set. Schiffer and Walther [122] developed a robust location routing approach that considers uncertain customer patterns, emphasizing the necessity of robust modeling to address the limitations of deterministic models in generating feasible solutions for all instances.

5.2. Solution approach

Fig. 12 presents the relationship between mathematical model (MM) and solution approach proposed by GVRP studies. Papers on GVRP studies are classified into two groups: those presenting a mathematical model formulation and those directly solving the problem with a proposed solution approach without formulating a mathematical model, classified as type 1. Among the 458 papers, 34 belong to this category. In type 1, some researchers define and describe the problem before presenting the proposed solution approach. Additionally, some researchers utilize mathematical models developed by others. For example, Londono et al. [123] introduced an iterated local search to solve the green vehicle routing problem with a private fleet and common carrier considering workload equity (GVRPPCWE). The mathematical model for the GVRPPCWE was derived from the formulation proposed by [124].

Among the 458 papers, 424 propose the formulation of a mathematical model for addressing the GVRP. These papers are categorized into three types: type 2, where researchers formulate and solve the MM solely using an exact solver; type 3, where researchers formulate the MM and solve the problem using the proposed solution approach; and type 4, where the MM is both formulated and solved using an exact solver and a proposed solution approach. Table 3 shows 44 papers classified in type 2. In type 2, the solution with the exact solver aims to address real case problems, validate the mathematical model, or evaluate the performance of the proposed model. For instance, Esmaili and Sahraeian [125] developed a MILP model for two-echelon capacitated vehicle routing problems with environmental considerations. They applied the proposed model to a real-world case in a supermarket chain operating in the Netherlands. Similarly, Tirkolaee et al. [92] formulated a MILP model for a multi-trip location-routing problem with time windows (MTLRP-TW) in medical waste management during the COVID-19 pandemic. A real-life case study was conducted in Sari city of Iran to assess the performance and applicability of the proposed model.

Table 3
List of GVRP studies classified as type 2.

Type of optimization	Authors	Quantity
Single-objective	Sholichah et al. [126], Leggieri and Haouari [68], Koyuncu and Yavuz [72], Kopfer and Vornhusen [78], Xiao et al. [86], Vakili et al. [90], Jharkharia and Das [110], Eshtehadi et al. [128], Khalafi and Zarei [127], Paz et al. [129], Qazvini et al. [130], Schiffer and Walther [131], Abdoli et al. [132], Bruglieri et al. [133], Shaohua et al. [134], Demir et al. [135], Kancharla and Ramadurai [136], Lin et al. [137], Madankumar and Rajendran [138], Mirmohammadi et al. [139], Murakami [140], Soysal et al. [141], Soysal et al. [142], Zuo et al. [143], Nasiri et al. [144], Cataldo-Díaz et al. [145], Hulagu and Celikoglu [146], Mahdavi et al. [147], Mulati et al. [148], Rathore et al. [149]	30
Multi-objective	Jabbarzadeh [91], Tirkolaee et al. [92], Parast et al. [116], Soleimani et al. [120], Toro et al. [150], Toro et al. [151], Ziaei and Esmaili and Sahraeian [152], Afshar-Bakeshloo et al. [153], Lerhlaly et al. [154], Tavana et al. [155], Stellingwerf et al. [156], Sun et al. [157], Defalque et al. [158], Sánchez et al. [159]	14
Total		44

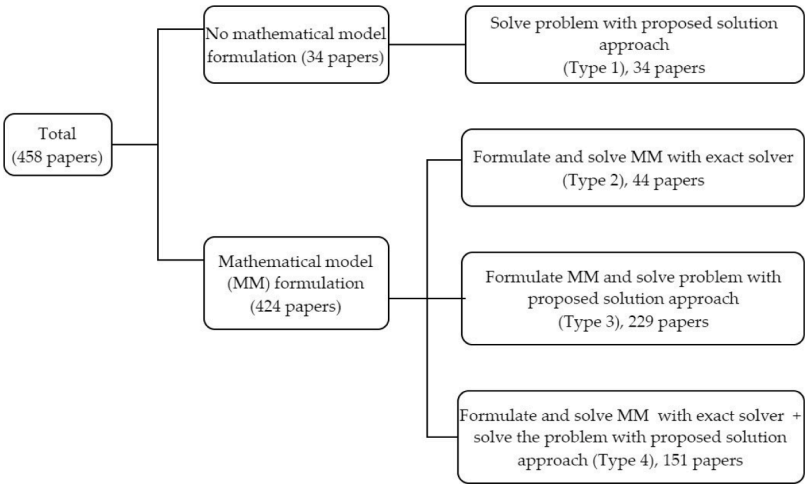


Fig. 12. Classification of papers on GVRP studies based on the relationship between mathematical models and solution approaches.

We also describe various previous studies in type 2 that used data from instance sets or randomly generated data. With these datasets, researchers employed the solution provided by the exact solver to validate or assess the performance of the proposed model. Zuo et al. [143] developed a MILP model for the electric vehicle routing problem with time window (EVRPTW) and conducted computational experiments using Solomon's VRPTW instances to validate their model. Bruglieri et al. [133] introduced a new MILP formulation for the GVRP, where the visits to the alternative fuel stations (AFSs) were implicitly considered. They evaluated their model's performance on two sets of benchmark instances sourced from [28]. Jharkharia and Das [110] developed a MILP incorporating the carbon cap and trade policy in vehicle routing with integrated order picking and delivery problems. They performed numerical experiments on various instances using randomly generated data. Soleimani et al. [120] proposed a multi-objective nonlinear programming model and tested its efficiency using a data simulation algorithm to create 10 small-and medium-scale problems for validation.

In type 3, researcher formulated MM and applied their proposed solution approach. Fig. 13 indicates that 229 papers fall into this category. For instance, Liu et al. [160] proposed a simulated annealing (SA) algorithm to optimize the joint distribution-green vehicle routing problem (JD-GVRP), demonstrating through real business data from four cold chain companies and 28 customers that joint distribution effectively reduces total costs and carbon emissions compared to single distribution. Similarly, Meng et al. [161] introduced a two-phase optimization approach to minimize both fuel consumption and emissions. Initially, their method utilized a fuzzy hierarchical clustering technique to group customers, followed by the development of an optimization model for the vehicle routing problem within each cluster, incorporating a genetic algorithm.

In type 4, researcher formulated MM and then solved it using both an exact solver and their proposed solution approach. According to Fig. 13, there are 151 papers included in this type. Typically, researchers employed two or three groups of problem instances (small, medium, and large-sized) to assess the performance of their proposed algorithm. For example, Tirkolaee et al. [162] generated 15 random instances of various sizes to validate their proposed mathematical model and evaluate the performance of the proposed algorithm. Results showed that their proposed HGA demonstrated suitable efficiency compared to SA and CPLEX. Additionally, Pitakaso et al. [115] demonstrated that all proposed algorithms significantly outperformed LINGO for larger problem sizes, although they produced results close to LINGO for small-sized instances. Nonetheless, for large-sized instances, there was a 7.77 % gap compared to the lower bound, highlighting potential computational time savings.

In total, 414 papers presented solution approaches for addressing MM, calculated by summing the papers in types 1, 3, and 4. Papers categorized as type 2 are labeled as having "no proposed approach". Additionally, some papers propose multiple solution methods, categorized as "additional solution approach". Fig. 13 provides insight into the distribution of various solution approaches for both single-objective and multi-objective GVRP.

In the context of single-objective GVRP, the hybrid approach demonstrates almost comparable performance to the metaheuristic approach between 2016 and 2023. However, regarding multi-objective GVRP, the metaheuristic emerges as the most frequently proposed method. Therefore, metaheuristics remained the preferred choice for addressing multi-objective GVRP. Our findings diverge slightly from [6], who indicated metaheuristic algorithms are dominant methods for all variants of GVRPs. This difference may be attributed to the timeframe considered in our review, extending up to 2023. Nevertheless, our results align with partial reviews of GVRP conducted by [8] and [22], indicating a notable increase in the adoption of hybrid algorithms for electric vehicle routing over the years. Consequently, the once-dominant metaheuristic approach for single-objective GVRP appears to be undergoing a shift towards hybrid approaches. Interestingly, according to Fig. 13, the utilization of hybrid M-M exceeded other hybrid categories, consistent with the results of [12] which indicated a higher prevalence of hybrid M-M.

Exact methods are more widely used to solve single-objective and multi-objective GVRP than the heuristic approach. One significant advantage of exact methods is their ability to guarantee optimal solutions. However, their usage is often limited by the size of instances due to the combinatorial nature of these problems. Currently, the exact method is often combined with heuristics or metaheuristics to obtain solutions of better quality. According to Fig. 13, 32 papers proposed a hybrid exact-metaheuristics (E-M) approach to solve GVRP. Furthermore, only 23 researchers proposed heuristics as a solution approach. The heuristic approach is rarely used by researchers to solve GVRP due to its tendency to yield suboptimal solutions and lower quality outcomes. To overcome these shortcomings, heuristics are combined with exact or metaheuristics to get better quality. 12 papers proposed a hybrid exact-heuristic (E-H) and 47 papers proposed a hybrid heuristic-metaheuristics (H-MH) approach to solve both single and multi-objective GVRP.

Given that a majority of papers in the combinatorial optimization problem focus on deterministic scenarios, it becomes imperative to explore simulation-optimization methods, which enable researchers and practitioners to address real-world models under uncertainty [163]. Hence, in this review, we introduce a new type of hybrid combination termed hybrid simulation optimization (S-O). As depicted in Fig. 13, seven papers employ this approach.

In our review, we have incorporated a new analysis for solution approach known as hyper-heuristics. The key distinction between metaheuristics and hyper-heuristics lies in their approach: while metaheuristics traditionally navigate through a problem solution's search space, hyper-heuristics concentrate on exploring a search space of heuristics. A hyper-heuristic is a heuristic search method that seeks to automate, often by the incorporation of machine learning techniques,

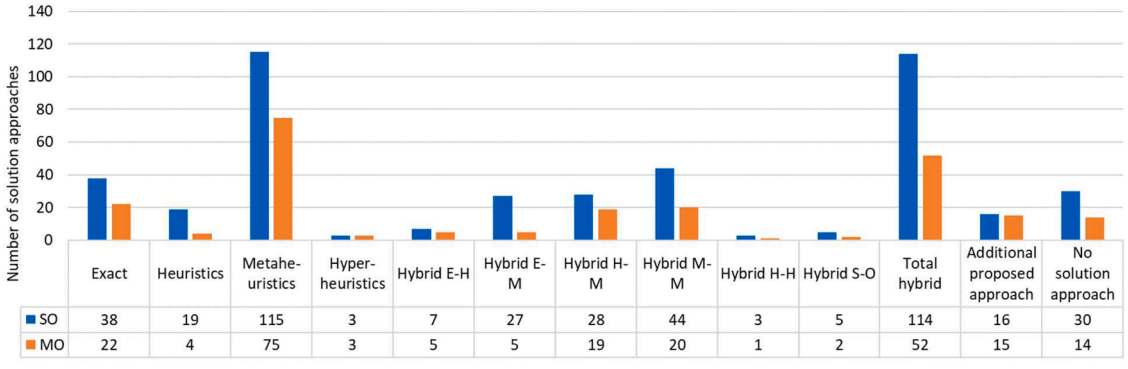


Fig. 13. The distribution of different solution approaches proposed for single and multi-objective GVRP.

the process of selecting, combining, generating, or adapting several simpler heuristics (or components of such heuristics) to efficiently solve computational search problems. The term "hyper-heuristics" emerged in 2000 to describe heuristics used for choosing heuristics in the context of combinatorial optimization [164]. Based on the results obtained from the review papers published between 2016 and 2023, we found that only 6 papers have employed the hyper-heuristics approach.

The present study highlights the usage pattern of solution approaches between 2016 and 2023. As depicted in Fig. 14, there is a significant increase in the application of metaheuristic algorithms in single-objective GVRP, although exceptions were observed in 2018 and 2021. Additionally, there was a increasing tendency in the adoption of hybrid algorithms until 2019, followed by a subsequent decline and stabilization until 2023. It is noteworthy that in 2016, 2018, 2019, and 2021, there were more researchers proposing hybrid algorithms compared to metaheuristics.

Similarly, Fig. 15 illustrates an increasing trend in the usage of metaheuristics for multi-objective GVRP, except for the years 2020 and 2021. In contrast, the utilization of hybrid algorithms has remained relatively stable over the past four years. While the number of papers employing metaheuristics consistently surpasses those utilizing hybrids, this pattern reversed in 2023. Lastly, researchers have initiated the use of hyper-heuristic algorithms to tackle GVRP since 2019, although such studies are still relatively scarce.

5.3. Software and programming language

Based on the explanation in subSection 5.1, there are 195 papers that proposed a mathematical model (MM) and solved the model with an exact solver approach. To accomplish this, researchers require a tool for modeling and solving MM. In Fig. 16, the general algebraic modeling system (GAMS) is the most commonly chosen programming language to express MM. GAMS is a high-level software for mathematical optimization, tailored for solving linear, nonlinear, and mixed-integer optimization problems. It includes an integrated development environment (IDE) and interfaces with various commercial and open-source solvers, such as CPLEX, Gurobi, BARON, and XPRESS. CPLEX and GUROBI solvers are predominantly employed by GVRP researchers for solving mixed-integer linear programming (MILP), while the BARON solver is used for addressing mixed-integer nonlinear programming (MINLP). Additionally, some other optimization programming language that are widely used to express MM are LINGO, IBM ILOG OPL, AMPL, and C/C++. Fig. 17 presents the most frequently used solvers, namely CPLEX, Gurobi, and LINGO. Among the several mentioned solvers, CPLEX is the most popularly used with a percentage of 70 %. In Fig. 16, 44 % of 195 papers did not provide information about the programming language used. Furthermore, only 8 % of 195 papers did not mention the solver used as shown in Fig. 17. It can thus be concluded that most authors/researchers report the solver without mentioning the programming language being used.

Based on the findings of the review, MATLAB is the most commonly used programming language for coding proposed solution approaches. Fig. 18 illustrates the distribution of programming languages, with MATLAB accounting for 32 % of usage, followed by C++ at 17 % and JAVA at 9 %. These three programming languages represent the most popular choices among researchers. Out of 137 GVRP studies utilizing MATLAB as the programming language, 74 were dedicated to single-objective GVRP solution approaches, whereas 63 were focused on multi-objective GVRP solutions. By comparing the studies considering multi-objective, which comprises only about 1/3 of the single-objective ones, it can be concluded that MATLAB is the most preferred programming language for addressing multi-objective GVRP. In addition, there are still some programming languages that are used such as Python, C#, C, VB, and GAMS but the percentage is relatively small. 20 % of 414 papers did not provide information about the programming language being used.

6. Proposed algorithm

To offer a more concise overview, the algorithms corresponding to each solution approach for both single and multiple-objective GVRP are summarized into tables. Many researchers present multiple algorithms to address GVRP with the aim of getting a better solution. SubSections 6.1 to 6.5 discuss the algorithms proposed for single-objective GCRP, while subSections 6.6 to 6.10 explore those proposed for multi-objective GVRP.

6.1. Exact algorithm for solving single-objective GVRP

Most researchers employed MILP to model the GVRP. For solving MILP model, the branch-and-cut, cutting plane, and branch-and-bound algorithm are commonly used [117]. The cutting plane and branch-and-bound techniques heavily rely on the simplex method multiple times in order to obtain the optimal solution with integer values. Branch-and-price builds upon the branch and bound framework by applying column generation throughout the branching and bound tree prior to branching.

From Fig. 13, it can be observed that 38 studies proposed exact approaches to solve single-objective GVRP. Further details regarding the algorithms developed by researchers in this approach can be found in Table 4. According to Table 4, researchers have developed a total of 38 exact algorithms for single-objective GVRP during the period of

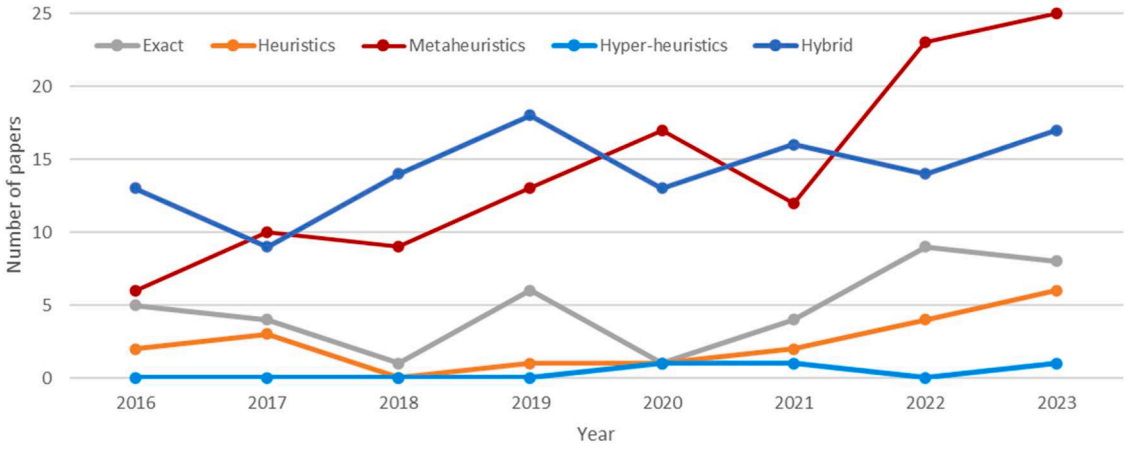


Fig. 14. The distribution of different solution approaches proposed in single-objective GVRP.

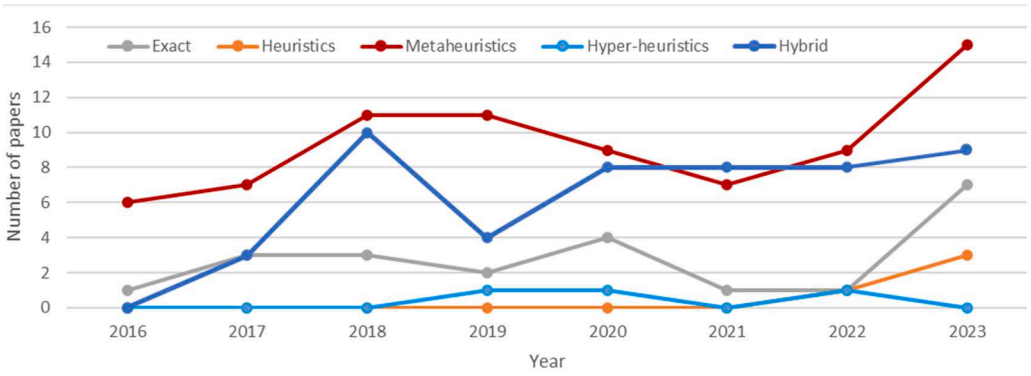


Fig. 15. The distribution of different solution approaches proposed in multi-objective GVRP.

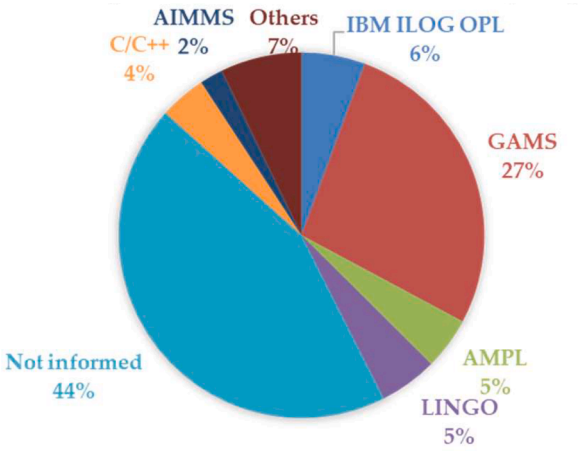


Fig. 16. Optimization programming language to express MM.

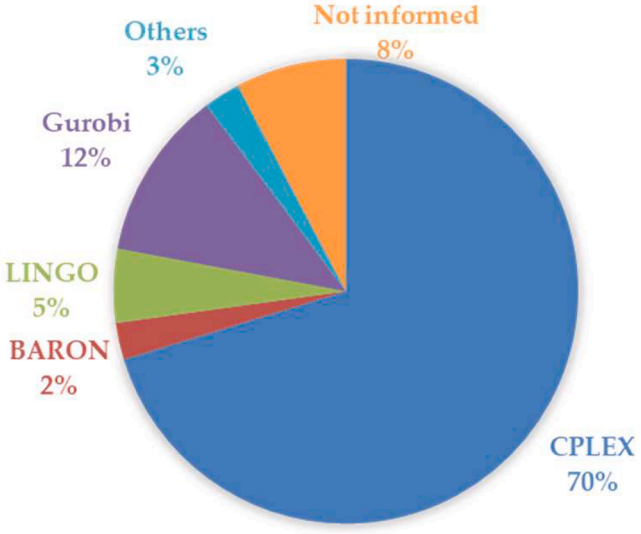


Fig. 17. Optimization solver to solve MM.

2016–2023. Among these, the branch-and-price (B&P) algorithm emerges as the most commonly utilized exact method for solving GVRP, followed by the branch-price-and-cut algorithm and branch-and-cut algorithm. Additionally, exact algorithms proposed by only one researcher are categorized under 'other exact algorithms'. More information about these algorithms can be found in Appendix 1.

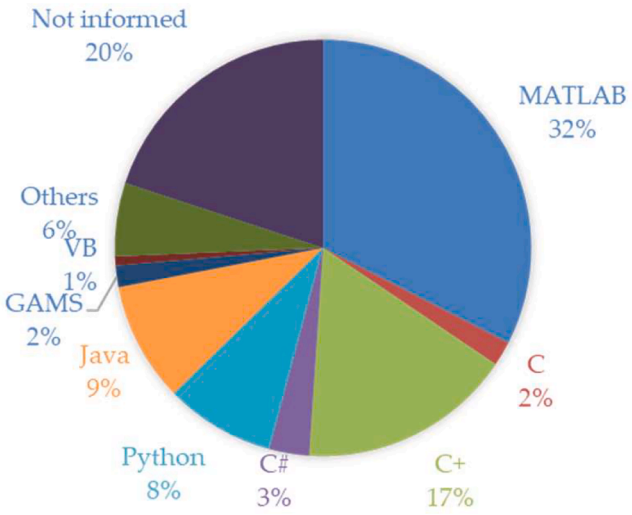


Fig. 18. Programming language for proposed algorithm.

Table 4
Exact algorithms proposed to solve single-objective GVRP.

Exact algorithm	Authors	Quantity
Branch-and-price	Yu et al. [42], Hiermann et al. [71], Qiu et al. [165], Dabia et al. [166], Behnke et al. [167], Wang et al. [168], Bezzi et al. [169], Purba et al. [170], Wang et al. [171]	9
Branch-price-and-cut/-based on a set-partitioning	Desaulniers et al. [64], Gao et al. [172], Luo et al. [173], Lam et al. [174], Subramanyam et al. [175]/Yin et al. [176]	6
Branch-and-cut	Cheng et al. [52], Ferreira et al. [177], Fazeli et al. [178], Yin and Zhao [179], Zhang et al. [180]	5
Cutting plane	Bruglieri et al. [57], Bruglieri et al. [181], Pelletier et al. [182]	3
Enhanced-/Logic-based Bender decomposition	Alkaabneh et al. [183], Attar et al. [184]	2
Other exact algorithms	Sun et al. [30], Yu et al. [185], Andelmin and Bartolini [186], Wang et al. [187], Yavuz [188], López-Castro et al. [189], Dukkanci et al. [190], Fukusawa et al. [191], Liu et al. [192], Shirzadi et al. [193], Chen et al. [194], Bruglieri et al. [195], Maheo et al. [196]	13
Total exact algorithms proposed for single-objective GVRP		38

Based on column generation approaches, efficient B&P algorithms can solve many GVRP variants comparatively quickly to reach optimality. The electric fleet size and mix vehicle routing problem with time windows and charging stations was presented by [71]. To obtain optimal solutions for smaller instances of their new benchmark and lower bounds for larger instances, they developed the branch-and-price method. Yu et al. [42] developed a method to quickly resolve the green vehicle routing problem for heterogeneous fleets with time windows. Based on their study, a multi-vehicle approximate dynamic programming algorithm was introduced to speed up the solution for the pricing problem. Then, a faster way to get a tighter upper bound is developed using the integer branch method. The results demonstrated that the modified B&P algorithm significantly decreased the branches and computational time. On a multigraph with load-dependent and vehicle-dependent arc weights, Behnke et al. [167] developed a column generation technique for solving an emission-oriented vehicle routing problem with heterogeneous fleets. Column generation is used to formulate an exact branch-and-price algorithm as well as a heuristic column generation approach.

6.2. Heuristic algorithm for solving single-objective GVRP

Based on the review findings, heuristic algorithms developed by researchers are highly specific to the GVRP problem and their GVRP models. As a result, researchers have given different names to the algorithms they have developed. According to Fig. 13, there are 19 single-objective GVRP studies proposing heuristic approaches. However, some studies propose more than one algorithm, resulting in a total of 24 heuristic algorithms, as shown in Appendix 2. For example, Alinaghian and Zamani [35] proposed three heuristic algorithms: green realistic opportunity saving (GROS-I), GROS-II, and green giant tour (GGT). Furthermore, Sweda et al. [197] developed two heuristic techniques for devising adaptive policies within a grid network. The initial heuristic generates multiple sample paths to rapidly assess the value of adaptive recharging decisions and chooses the optimal path, whereas the second heuristic incorporates further adaptability in path selection to enhance performance.

Heuristics are techniques for learning and solving problems based on experience [198]. These techniques aim to find solutions based on specific problem knowledge. While it does not guarantee an optimal solution, heuristics offer satisfactory solutions within a reasonable computational time. In the context of VRP research, heuristic techniques can be categorized into two types: constructive heuristics and improvement heuristics [8]. Constructive heuristics are often used to generate an initial solution via either serial or parallel route construction. On the other hand, improvement heuristics or local search (LS) explore the neighborhood around the current solution in seek of a better solution. The nearest neighbor, giant tour, and Clarke-Wright saving algorithm are a few well-known constructive heuristics employed in GVRP. Researchers that have proposed constructive heuristic algorithms to solve the single-objective GVRP are outlined below:

1. Alinaghian and Zamani [35] developed a novel heuristic method called Green Realistic Opportunity Saving (GROS)-I to deal with fleet size and mix vehicle routing problem to minimize fuel consumption. They modified the Realistic Opportunity Saving (ROS) developed by Golden et al. (1984) for GVRP.
2. Alinaghian and Zamani [35] proposed an algorithm named the Green Giant Tour (GGT) algorithm. A giant tour is constructed in the first step to visit every customer. In the second stage, the giant tour is divided up into sub-tours of which the start and end points are originated.
3. Solano et al. [199] developed a greedy algorithm to solve VRP with simultaneous pickup and delivery (VRPSPDTW). This heuristic used a time matrix as the main reference and the nearest neighbor algorithm to visit different customers.
4. Saka et al. [200] calculated a cost difference estimation for each potential move. With a given solution and cost change estimations for various moves, it is possible to select the most favorable moves. They demonstrated that this could enhance the efficiency of a local search heuristic. Additionally, they developed an exact solution heuristic in which the neighborhood is constructed using real cost change data rather than estimation.
5. Almouhanna et al. [201] proposed MSBRH algorithm which consists of two stages. In the first stage, several depots are selected to be opened from the list of potential candidates. In the second stage, customers are assigned to the selected depots and an initial routing solution is generated using Extended Savings Heuristic (ECWH) proposed by [202].

Researchers that proposed improvement heuristic algorithms to solve the single-objective GVRP include:

1. Alinaghian and Zamani [35] proposed GROS-II. The difference with GROS-I lies in two aspects: (1) the selection of pair of the nodes considering the saving and using roulette wheel selections, not based on maximum saving, (2) using local search 2-opt to optimize modified sub-tours.
2. Almouhanna et al. [201] proposed MSBRH algorithm. The best solutions found during the first stage are then improved in the second stage, which consists of two levels. In the global level, a biased-randomized version of ECWH is applied to the initial solution resulted from the first stage. Whereas in the local level, a biased-randomized version of the classical savings heuristics is employed to improve the routes associated with each depot.

6.3. Metaheuristic algorithm to solve single-objective GVRP

Table 5 provides an overview of the metaheuristic algorithms utilized in addressing single-objective GVRP. From 2016 to 2023, researchers proposed a total of 127 metaheuristic algorithms for GVRP. However, the number of studies proposing metaheuristic solution approaches during 2016–2023 amounted to 115, as indicated in Fig. 13. Hence, it can be inferred that a considerable number of single-objective GVRP studies proposed more than one algorithm.

Various metaheuristic classifications have been proposed based on exploration, exploitation, and search procedures. However, the dominant classification categorizes metaheuristics into trajectory-based (single-based) and population-based, as well as nature-inspired and non-nature-inspired [203]. This classification, which distinguishes between single-based (S-metaheuristics) and population-based (P-metaheuristics), will be utilized in analyzing previous GVRP studies. Out of those 127 algorithms, 67 fall into the category of S-metaheuristics. Among those S-metaheuristic algorithms, adaptive large neighborhood search (ALNS) was the predominantly used S-metaheuristic algorithms for single-objective GVRP, succeeded by simulated annealing (SA), and variable neighborhood search (VNS). Subsequently, 60 out of 127 proposed P-metaheuristic algorithms during 2016 to 2023. Ant colony optimization (ACO) emerged as the most employed P-metaheuristic algorithms for single-objective GVRP, followed by genetic algorithm (GA) and particle swarm aptimization (PSO). Population-based metaheuristics are associated with evolutionary algorithms (EA) and swarm intelligence (SI) algorithms. Within the EA category, various well-known algorithms exist, such as genetic algorithm and differential evolution. Swarm intelligence algorithms have received less attention, with only a few studies focusing on their application. For instance, bee colony optimization, camel algorithm, firefly algorithm, and intelligent water drop were each proposed by a single researcher, as indicated in Appendix 3.

Researchers made different modifications and improvements in terms of parameters or randomization operators to increase the convergence of the algorithm and the solution, which is indicated by the

Table 5
Metaheuristic algorithms proposed to solve single-objective GVRP.

Metaheuristic algorithm	Category	Authors	Quantity
Improved- /Oriented-/Two-stage-amplified-/Three-dimensional-/ACO	P	Liu et al. [100], Fan et al. [204], Fan [205], Li et al. [206], Yao et al. [207]/ Zhang et al. [208]/Kumar et al. [209]/Guo et al. [210]./Zhang et al. [211], Elasseri et al. [212], Su and Fan [213], Meng and Ma [214], Adhitama and Kurniawati [215], Stamadianos et al. [216]	14
Adaptive-/ Cycle Evolutionary-/ Improved-/Genetic Algorithm	P	Wang and Wen [41]/ Wang et al. [217]/ Gao and Ye [218], Liu and Zhang [219]/Hsueh [220], da Costa et al. [221], Moryadee et al. [98], Sruthi et al. [222], Zhu and Wen [223], Foroutan et al. [44], Moazzeni et al. [224]	11
Improved-/ Self-Adaptive Learning-/Chaotic-/ Two-stage search quantum-/Adaptive-/PSO	P	Naderipour and Alinaghian [225]/ Srijaroon et al. [226]/Zhang et al. [227] /Zhao et al. [228]/Ding et al. [229]/Hannan et al. [230], Baiq and Garside [231]	7
Diversity-enhanced- /Memetic algorithm/-with competition	P	Xiao et al. [232]/ Peng et al. [60], Mara et al. [233], Shi et al. [234]/ Wang and Lu [61]	5
Differential evolution	P	Karami and Ghezavati [118], Kunnapapdeelert and Klinsrisuk [104], Alinaghian et al. [235]	3
Improved-/Firefly algorithm	P	Alinaghian and Naderipour [236]/Ramachandranpillai and Arock [237], Behnamian et al. [238]	3
Ant colony system	P	Zhang et al. [239], Kyriakakis et al. [240]	2
Cuckoo search algorithm	P	Ng et al. [48], Pratap et al. [241]	2
Enhanced-/ Improved chemical reaction optimization	P	Poonthalir and Nadarajan [242], Pu et al. [243]	2
Max–min ant system (MMAS)	P	Stamadianos et al. [216], Kyriakakis et al. [240]	2
Other P-metaheuristic algorithms		Hatami et al. [36], Stamadianos et al. [216], Pratap et al. [241], Liu et al. [244], Chen et al. [245], Utama et al. [246], Quan et al. [247], Thakur et al. [248], Rahmanifar et al. [249]	9
Total P-metaheuristic algorithms proposed for single-objective GVRP			60
Improved-/Extended-/Robust-/Template-based-/Adaptive large neighborhood search/-based heuristic	S	Koç et al. [250], Keskin and Çatay [251], Alinaghian et al. [252], Shi et al. [253], Zhang et al. [208]/ Zhang et al. [254]/Wen et al. [255]/ Nolz et al. [256]/ Kancharla and Ramadurai [38], Majidi et al. [111], Schiffer and Walther [131], Franceschetti et al. [257], Majidi et al. [258], Zhang et al. [259], Koç [260], Yu et al. [261], Xu et al. [262], Liu et al. [263], Ma et al. [264], Dönmez et al. [265], Erdem [266]/Erdem et al. [267]	22
Competitive-/Improved-/simulated annealing/-with a restart strategy/clustering-based SA with temperature-dependent penalty cost function	S	Dai et al. [268]/Feng et al. [269]/ Foroutan et al. [44], Abdoli et al. [59], Sherif et al. [119], Liu et al. [160], Rahmanifar et al. [249], Li et al. [270], Kazemian and Aref [271], Eydi and Alfi [272], Normasari [273], Nosrati and Khamseh [274], Reinaldi et al. [275], Elahi and Darestani [276], Lijun et al. [277]/ Yu et al. [67]/ Ibnoulouafi et al. [278]	17
Adaptive-/ Biased-Randomized-/General -/ Improved -/Iterated -/ Parallel-/Variable neighborhood search/- adaptive Search	S	Hof et al. [279]/ Almouhanna et al. [201]/ Affi et al. [56], Karakostas et al. [280]/ Matijević [281]/Ren et al. [282], Chen et al. [283]/Lu et al. [284]/ Zhou et al. [285]/Rezgui et al. [75], Chen et al. [286]/ Pitakaso et al. [115]	12
Augmented-/Split-based Adaptive-/Iterated-/Tabu Search	S	Alinaghian et al. [235]/ Li et al. [49]/ Subramanyam et al. [175]/ Moryadee et al. [98], Solano et al. [199], Zhang et al. [208], Alinaghian et al. [235], Ehmke et al. [287], Setak et al. [288], Nosrati and Khamseh [274]	10
Multi-start-/Iterated local search	S	López-Castro et al. [188]/ Macrina et al. [79], Dukkanci et al. [189], Granada-Echeverri et al. [289]	4
Other S-metaheuristic algorithms		Ren et al. [290], Bruglieri et al. [291]	2
Total S-metaheuristic algorithms proposed for single-objective GVRP			67
Total metaheuristic algorithms proposed for single-objective GVRP			127

P: P-metaheuristics, S: S-metaheuristics.

words "adaptive" and "improved" in the front or back of the basic metaheuristic algorithms. Some examples of this are improved simulated annealing (ISA) and adaptive genetic algorithm (AGA). Feng et al. [269] proposed ISA to solve VRP where travel speed is considered to be stochastic, with the aim of minimizing fuel consumption. They used a self-adaptive method to obtain the best initial temperature of the algorithm. High initial temperature increases the computational time while low initial temperature increases the risk of getting trapped in a local optimal solution. Therefore, they generated the maximum temperature adaptively according to the problem; that is, by randomly changing the incumbent solution 1000 times. Wang and Wen [41] proposed an adaptive genetic algorithm for optimizing the low-carbon two-echelon heterogeneous-fleet vehicle routing problem (LC-2EHVRP). They employed an adaptive crossover and mutation mechanism with an adaptive mechanism to control the values of the probabilities of crossover (P_c) and probabilities of mutation (P_m).

6.4. Hyper-heuristic algorithm to solve single-objective GVRP

Hyper-heuristics (HH) are often considerably more effective than heuristic or metaheuristics when applied to a specific problem. This is primarily due to its ability identify and combine the best traits of individual low-level heuristics [292]. The core idea of hyper-heuristics is to solve optimization issues using a variety of simple and flexible low-level heuristics (LLHs), as well as to provide a framework for controlling the selection and application of LLHs. Across the search space of heuristics, high-level heuristic and metaheuristics are typically utilized as search methods.

According to Sánchez et al. [24], the popularity of hyper-heuristics research has risen in several most relevant problem fields. Sánchez et al. [24] concluded the following combinatorial optimization problems to experience the top six citation growth: (1) job-shop scheduling, (2) vehicle routing, (3) production scheduling, (4) traveling salesman, (5) set covering, and constraint satisfaction. Although Sanchez et al. [24] noted that despite the popularity of VRP in recent years, there are still very few researchers who have utilized hyper-heuristics to solve GVRP. Zhang et al. [293], Olgun et al. [294], and Shang et al. [295] focused on utilizing hyper-heuristics for addressing single-objective GVRP. Zhang et al. [293] proposed HH to solve time-dependent green location routing problem with time windows. Olgun et al. [294] proposed a hyper-heuristic based on iterative local search (HH-ILS) to solve GVRP with simultaneous pickup and delivery (G-VRPSPD). To prove the effectiveness of the HH-ILS, comparative tests were conducted on instances of the classic VRPSPD. When compared to various heuristics previously proposed in the literature, their algorithm produced 38 best known solutions on Dethloff instances and 5 best known solutions on Salhi and Nagy instances.

6.5. Hybrid algorithm to solve single-objective GVRP

Hybrid algorithms are sets of two or more algorithms that work together and complement one another to produce a beneficial synergy [7]. Sometimes using a specific approach result in disadvantages, such as low-quality solutions, getting trapped in local optima in the search space, and high computation time. As a result, researchers carefully combine two or more algorithms to utilize the benefits of each algorithm simultaneously [6].

In this section, we focus on some recent hybrid algorithms that have been implemented on a wide range of single-objective GVRP. Since there is no standardization of naming, some researchers adopted very different names for their hybrid algorithms. Therefore, in this review, we do not mention the name of their hybrid algorithms. Instead, we provide additional details regarding the combination of algorithms employed by researchers as the basis for the proposed hybrid algorithm. Hybrid algorithms leverage the strengths of individual algorithms to enhance solution quality and efficiency. Understanding the combination of algorithms provides insights into how these algorithms complement each other and work synergistically to tackle the challenges posed by the green vehicle routing problem.

6.5.1. Hybrid exact-metaheuristic algorithm for single-objective GVRP

The combination of mathematical programming (MP) and metaheuristics offers a powerful and flexible methodology to solve complex optimization problems. This hybridization of MP with metaheuristics leads to the matheuristic approach. Matheuristics aim to leverage the strengths of both MP, which provides rigorous and exact optimization solutions, and metaheuristics, which offer efficient and approximate solutions. The availability of efficient optimization software, libraries, and frameworks for MP and high-level modeling languages is expected to lead to a growth in matheuristic approaches [296].

As shown in Table 6, there are 27 hybrid exact-metaheuristic algorithms proposed by GVRP researchers. This number corresponds exactly to the number of single-objective GVRP studies proposing hybrid exact-metaheuristic approaches (as indicated in Fig. 13). Futhermore, S-metaheuristics emerge as the predominant algorithms when combined with exact algorithms. From S-metaheuristic algorithms, large neighborhood search (especially ALNS), VNS, and ILS are the top three algorithms most commonly used. Meanwhile, among the P-metaheuristic algorithms proposed by GVRP researchers, only GA, KSA, and memetic algorithm have been employed, each with only one study. Additionaly, mixed integer programming (MIP) stands out as the most common exact method combined with metaheuristic algorithms.

Talbi [296] proposed a taxonomy outlining hybridization strategies that integrate exact and metaheuristic algorithms. This taxonomy is similar to the classification suggested by [7], which also categorizes hybridization based on whether the algorithm is integrated directly or collaboratively with a metaheuristic algorithm. Subsequently, we will discuss in more detail the combination of exact and metaheuristic algorithms, drawing upon the taxonomy proposed by [296].

1. Low-level relay hybrid (LRH). LRH represents hybrid schemes in which S-metaheuristics is embedded into an exact approach, or vice versa, to improve the search strategy.
   a) Metaheuristics have the capability to address various search problems encountered during the development of exact methods, encompassing tasks such as node selection strategy, upper bound generation, and column generation. Koç and Karaoglan [55] introduced S-metaheuristics integrated into an exact algorithm, presenting a new solution method combining simulated annealing and branch-and-cut algorithm to address G-VRP. Simulated annealing enhances the initial solution and the upper bounds identified during the search process.
   b) This hybrid approach integrates exact algorithms into the exploration components of S-metaheuristics. Shao and Dessouky [63] addressed the vehicle routing problem for alternative fuel vehicles with fixed fueling time. They proposed a hybrid heuristic approach combining ALNS with local search and MILP model to handle large-scale problems. The ALNS iterative process includes local search to intensify exploration, followed by destruction and repair, fleet size minimization, and fueling plan optimization using the MILP model.

1. Low-level teamwork hybrid (LTH). In LTH, a search component of P-metaheuristic is replaced by another optimization algorithm. Xiao and Konak [307] proposed a hybrid solution approach that integrates dynamic programming into genetic algorithm.
2. High-level relay hybrid (HRH). This hybrid algorithm employs separate algorithms sequentially. Initially, a metaheuristic is used to reduce the problem size by techniques like partitioning decision variables into sets X and Y. The metaheuristic optimizes variables in set X, while an exact method solves the remaining set Y. Keskin et al. [297] proposed a matheuristic algorithm combining ALNS and MILP to solve EVRP with time-dependent waiting times at recharging stations. ALNS finds feasible routes, while MILP optimizes charging decisions. Additionally, Palletier et al. [182] introduced a two-phase heuristic for large instances of the electric vehicle routing problem with energy consumption uncertainty (EVRP-ECU). The first phase constructs routes using the LNS framework, and in the second phase, a set partitioning (SP) problem generates a complete robust solution from the routes pool.

Table 6
The combination of algorithms proposed in the hybrid exact-metaheuristic for single-objective GVRP.

Metaheuristic algorithm	Exact algorithm							Total
	Mixed-/Integer programming	Set partitioning	Branch and cut	Column generation	Dynamic programming	POM	Set co-vering	
A-/LNS	[63,66,82,297,298,299]	[65,182,300]		[301]	[122]			11
VNS	[40,302,303,304]							4
ILS		[74,305]					[306]	3
GA					[307]			1
GRASP						[84]		1
KSA	[308]							1
LS & TS			[309]					1
MA	[310]							1
SA			[55]					1
SA & GRASP	[94]							1
TS				[311]				1
VND			[312]					1
Total	13	5	3	2	2	1	1	27

KSA: Kernel search algorithm, MA: memetic algorithm, GRASP: Greedy randomized adaptive search procedure POM: Partial optimization. method.

3. High-level teamwork hybrid (HTH). HTH is a parallel cooperative approach combining metaheuristics algorithms and MP. A key challenge in HTH hybrid is exchanging information between these components. Metaheuristics algorithms and MP solve different aspects of the problem and can exchange relevant information to enhance the hybrid approach. Schiffer and Walther [122] developed a parallelized hybrid of ALNS and DP. A corridor-based approach is used to guide local search and the DP component, intensifying the search in promising areas of the search space.

6.5.2. Hybrid exact-heuristic algorithm for single-objective GVRP

Table 7 indicates that there are 7 hybrid exact-heuristic algorithms proposed in single-objective GVRP research. As mentioned earlier, the utilization of heuristics has declined due to its limitations when compared to metaheuristics. This trend is also reflected in the study of single-objective GVRP, where the quantity of hybrid exact-heuristic algorithms comprises only a quarter of the total hybrid exact-metaheuristic algorithms.

Mixed integer programming (MIP) stands out as the most common exact method combined with heuristic algorithms. Among the heuristic algorithms employed are the greedy algorithm, multi-start local search, and sweep algorithm. The greedy algorithm is widely regarded as the most prominent heuristic in literature, owing to its several advantages. These include its simplicity in both comprehension and implementation, as well as its rapid computation time. For instance, Cárdenas-Barrón and Melo [313] addressed the selective and periodic inventory routing problem (SPIRP) for waste vegetable oil collection by applying a fast and effective MIP-based heuristic. This heuristic aids in route construction, with one of its stages utilizing the greedy algorithm for the traveling salesman problem. Similarly, Chaabane et al. [314] also utilized the greedy heuristic for route construction.

6.5.3. Hybrid heuristic-heuristic algorithm for single-objective GVRP

From the results of the GVRP review, it can be concluded that the hybrid of two heuristic algorithms is the least proposed solution approach compared to other hybrid types. Based on the review, we identified three researchers who used a combination of two heuristic algorithms to solve the single-objective GVRP. Zhang et al. [239] proposed a two-phase heuristic for the capacitated GVRP (CGVRP). A Traveling Salesman Problem (TSP) is first solved by using the Nearest Neighborhood, and the solutions found in phase one are used to build the solutions of the CGVRP. When building the CGVRP route, the remaining fuel and products in the vehicle are calculated whilst inserting a customer. Futhermore, Li et al. [320] proposed two-stage heuristic algorithm. The first stage constructs the vehicle routes using the Clarke and Wright savings algorithm (CW). In the second stage, the feasible solution is improved by the use of a local search. In this stage, relocate and λ-interchange operators were proposed for the search for neighborhood solutions. In addition, Rao et al. [321] proposed an efficient bi-objective hybrid local search algorithm. They used a hybrid of three local search heuristics to generate optimal capacitated vehicle routing problem solutions in the first stage, namely petal, nearest neighborhood, and insertion algorithms. In the second stage, they locally adjust the solutions from the first stage to improve them towards the direction of reducing fuel consumption.

Table 7
The combination of algorithms proposed in the hybrid exact-heuristic for single-objective GVRP.

Heuristic algorithm	Exact algorithm			Total
	MIP	Lagrangian relaxation	Branch-and-bound	
Greedy algorithm	[313, 314,315]		[316]	4
Alternating direction method of multipliers		[317]		1
Multi start local search	[318]			1
Sweep and 2-opt algorithm		[319]		1
Total	4	2	1	7

6.5.4. Hybrid metaheuristics-metaheuristics algorithm for single-objective GVRP

Table 8 presents the combination of metaheuristics with complementary metaheuristics proposed for addressing single-objective GVRP. Based on the findings of the review, this type of combination is more prevalent than the combination of metaheuristic algorithm with exact method, heuristic algorithm, or simulation. As shown in Table 8, there are 50 hybrid metaheuristic-metaheuristic algorithms proposed by single-objective GVRP researchers. This number corresponds greater than the number of hybrid exact-metaheuristic approaches, which is 44 (as shown in Fig. 13). This indicates that out of the 44 studies proposing this approach, some studies propose more than one hybrid exact-metaheuristic algorithm. For example, Masmoudi et al. [51] proposed three population-based hybrid artificial bee colony for solving the heterogeneous fleet vehicle routing problem with synchronized visits (HF-VRPS). Kyriakakis et al. [240] introduced the electric vehicle routing problem with drones (EVRPD) for package delivery. To enhance solution quality, a variable neighborhood descent (VND) algorithm is included as a local search procedure. Within this framework, hybrid ant colony system (HACS) and hybrid min-max ant system (HMMAS) are developed to address the EVRPD.

VNS and its variations, GA, SA, and TS are metaheuristic algorithms commonly proposed by researchers for combination with other metaheuristic algorithms. According to Table 8, VNS and its variations are most frequently combined with GA, PSO, and TS. Similar to the hybrid exact-metaheuristic approach, single-based metaheuristics (including VNS and its variations, SA, TS, LNS and its variations, and VND) are often chosen to complement other metaheuristic algorithms. In this review, the hybridization classification of these combinations aligns with the taxonomy suggested by [296]:

1. Low-level relay hybrid (LRH). This class of hybrids represents algorithms where any given metaheuristic algorithm is embedded into the S-metaheuristic algorithm. Molina et al. [322] introduced the VNTS algorithm, embedding the TS algorithm into the S-metaheuristic algorithm to improve the basic VNS algorithm. VNTS algorithm is presented to effectively reduce the pollutant emissions in an Eco-WCVRP.
2. Low-level teamwork hybrid (LTH). LTH algorithms integrate an S-metaheuristic into P-metaheuristics. This hybrid type is widely favored and has proven to be successful in addressing various optimization problems. Cao et al. [43] developed a hybrid method by merging LNS and GA algorithms. In this approach, the destruction and repair mechanism of LNS is employed to improve the local search capability within GA. Erdem and Koç [323] introduced a local search technique utilizing the VND algorithm, which utilizes swap and relocation neighborhoods, to enhance the population in GA.
3. High-level relay hybrid (HRH). HRH algorithms execute self-contained metaheuristics sequentially, wherein the initial solution of one metaheuristic may be produced by another optimization algorithm. Abedinzadeh et al. [112] utilized GA to obtain an initial solution, followed by SA to enhance the generated solutions. Utama et al. [324] suggested enhancing the Butterfly optimization algorithm (BOA) by adjusting 10 % of search agent positions.
4. High-level teamwork hybrid (HTH). The HTH structure involves a number of different algorithms performing a parallel search and working together to find an optimal solution. Lyu and He [335] developed a two-stage hybrid metaheuristic approach (TSHM), utilizing an improved GRASP in the first stage to generate an initial feasible solution, followed by a hybrid genetic algorithm (HGA) in the second stage to improve the initial population.

Table 8
The combination of algorithms proposed in the hybrid metaheuristics-metaheuristics for single-objective GVRP.

Metaheu-ristic algorithm	A-/G-/VNS	SA	S-/VND	AL NS	PSO	GRASP	AMP	BOA	DA	ELS	ICA	ILS	OBA	RRT	TA	TS	VSD	WOA	Total
GA	[102,178,327,328,329,330]	[37,112,162,331,332]	[323]	[43]	[333]	[334]				[335]					[336]	[337]			18
TS	[62,322]	[47,338]			[339, 340]			[324]										[341]	8
ACO	[73,212,342]			[343]								[344]							5
ABC							[345]		[51]				[51]	[51]					4
A-/LNS	[346]	[347]	[348]			[183]													4
PSO	[81,349,350]			[351]															4
A-/VNS				[352]							[121]								2
ILS			[353]														[354]		2
ACS			[240]																1
MA			[70]																1
MMAS			[240]																1
Total	15	8	6	4	3	2	1	1	1	1	1	1	1	1	1	1	1	1	50

AMP: Adaptive memory programming, OBA: Old bachelor algorithm, DA: Demon algorithm, ELS: Elite neighborhood search, ICA: Imperialist competitive algorithm, MA: Memetic algorithm, SVND: Sequential variable neighborhood descent, RRT: Record-to-record travel algorithm, VSD: Variable search descending, TA: Threshold accepting.

6.5.5. Hybrid heuristic-metaheuristic algorithm for single-objective GVRP

According to Table 9, single-objective GVRP researchers have proposed 28 hybrid heuristic-metaheuristic algorithms, which corresponds precisely to the number of hybrid exact-metaheuristic approaches depicted in Fig. 13. Heuristics are incorporated into metaheuristic algorithms to enhance local search exploration. Local search heuristics refine solutions through minor adjustments to routes, considering factors such as neighborhood structures or local optimization techniques. This combination enhances the exploitation capability of metaheuristics, leading to improved solutions. Commonly proposed heuristic algorithms include local search and Clark Wright saving (CW). Additionally, GA is the preferred metaheuristic algorithm for combination with heuristic algorithms in solving single-objective GVRP.

6.5.6. Hybrid simulation-optimization algorithm for single-objective GVRP

The advancement in computational capabilities has led to the emergence of methods that integrate simulation and optimization, bridging the gap between these traditionally separate approaches in the field of operational research [325]. Hybrid simulation-optimization methods aim to tackle the challenge of uncertainty, which can be addressed through various conventional methods like stochastic programming, fuzzy programming, and stochastic dynamic programming. To address this challenge, there is a growing trend to combine metaheuristics with other solution approaches, particularly simulation, to effectively manage real-life uncertainty characterized by stochastic or random behavior. This integration, known as "simheuristics," was introduced by [163]. Simheuristics involve designing a hybridization wherein the simulation component deals with the model's uncertainty and interacts with the metaheuristic component, which explores the solution space to find a near-optimal solution for real-world optimization problems. Table 10 shows the proposed simulation-optimization algorithm to solve the single-objective GVRP. Among these papers, Reyes-Rubiano et al. [326] addresses the consideration of driving-range constraints and stochastic travel times to design realistic electric vehicle routing plans. They propose a simheuristic algorithm that combines Monte Carlo simulation with multi-start biased-randomization techniques.

Based on the metaheuristic algorithms most frequently proposed in Tables 5-6 and 8-10, the summaries are presented in Table 11. The top three algorithms proposed by researchers to address single-objective GVRP studies are LNS and its variants (especially ALNS), GA, and VNS with its variants (especially AVNS). According to [379], heuristics based on LNS and its variant have demonstrated significant success in solving various scheduling and transportation problems. The single-objective GVRP review findings further support this statement. One of the key advantages of LNS heuristics is their ability to quickly assemble a heuristic using existing components. It is possible to transform an existing construction heuristic or exact method into a repair heuristic, and the implementation of a destroy method based on random selection is straightforward [379].

Moghdani et al. [6] concluded that the most commonly employed metaheuristic methods, in order, are GA, SA, ALNS, and TS. Our review findings reveal a slightly different ranking of algorithms compared to the findings of [6]. Furthermore, we identify VNS, ACO, and PSO as frequently used algorithms in addressing single-objective GVRP. The top three most employed metaheuristic algorithms are LNS and its variants (especially ALNS), GA, and VNS along with its variants. Recognizing these algorithms as the predominant algorithms in single-objective GVRP research between 2016 and 2023 offers valuable insights into established methodologies, benchmarks for comparison, opportunities for addressing common challenges, potential for hybridization, and practical implications for real-world applications.

6.6. Exact algorithm for solving multi-objective GVRP

A multi-objective optimization problem (MOP) consists of several

Table 9
The combination of algorithms proposed in the hybrid heuristic-metaheuristics for single-objective GVRP.

Metaheu-ristics	Heuristic algorithm																	Total
algorithm	LC	NS	CW	KCS	KM	H1-H9	DM	FC	Gr	HC	IMWMTT	PO	QRG	RR	SD TOA	SO-MTO	SPA	TRA	
GA	[355,356]							[161]	[357]	[358]		[359]		[360]				[361]	8
A-/LNS	[71,362]		[363]												[364]	[365]	[366]		6
ACO				[367]	[367]						[368]								3
HNN				[369]	[369]														2
TS		[370,371]																	2
ALNS & IG			[372]																1
FA							[238]												1
PSO	[373]																		1
SA						[249]													1
SEO						[249]													1
QI													[374]						1
VNS		[375]																	1
Total	5	3	2	2	2	2	1	1	1	1	1	1	1	1	1	1	1	1	28

FC: Fuzzy clustering, LC: Local search, PO: Post-optimization, TRA: Trust region algorithm, NS: Neighborhood search, IH: Improvement heuristics, HC: Hill climbing, QRG: Quantum rotation gate, QI: Quantum immune, KCS: K-means clustering, KM: K-Medoids, IMWMTT: Improved MWMTT, DM: Data mining FA: Firefly algorithm, RR: Ruin and recreate; SEO: Social engineering optimizer, H1-H9: Heuristics 1 to 9 developed by [249], HNN: Hopfield neural network, SPA: Shortest path algorithm, IG: Iterative greedy, SO-MTO:Speed optimization-multi-path traffic-covering optimization, Gr: Greedy algorithm, SDTOA: Speed-and-departure-time optimization algorithm.

Table 10
The combination of algorithms proposed in hybrid simulation-optimization for single-objective GVRP.

	Optimization algorithm			Quantity
	Multi-start biased-randomized	Genetic algorithm	Restricted dynamic programming	
Simulation	[326]	[376]	[95,377,378]	5

Table 11
Summary based on metaheuristic algorithms frequently employed in solving single-objective GVRP.

Metaheuristic	Metaheuristic	Most frequently proposed algorithms based on the Table					Total
algorithm	category	5	6	8	9	10	
LNS and its variants	S	22	11	8	7	–	48
GA	P	11	1	18	8	1	39
VNS and its variants	S	12	4	19	1	–	36
SA	S	16	2	8	1	–	27
TS	S	10	2	9	2	–	23
ACO	P	14	–	5	3	–	22
PSO	P	7	–	7	1	–	15

conflicting objectives that must be minimized or maximized. The nature of these objectives often leads to a trade-off, where improving one objective may come at the expense of worsening another. As a result, finding a single solution that optimizes all objectives is challenging. Instead, it is common to seek a set of solutions based on the non-dominance criterion. A solution that is not dominated by other solutions are called non-dominated solution or Pareto-optimal solution. Multi-objective optimization methods can be classified into three categories: scalarization technique, Pareto methods, and non-scalar and non-Pareto methods. Scalarization aggregates the objective functions (or reformulate them as constraints) and solved the problem a single-objective problem [380]. There are many different types of scalarization algorithms that solve the MOO problem. The weighted sum method is the simplest and most popular scalarization method. Given a set of weights for each objective, the multi objectives are combined into a single objective by a weighted sum. On the other hand, the epsilon (ε) constraint method works by minimizing the most preferred objective function while considering the remaining objective functions as constraint bounds using the parameter ε_j [381]. The weakness of this method is that the ε_j vector must be chosen carefully so that it is within the minimum or maximum values of the individual objective function. However, this method is applicable for either convex or non-convex problems, making it a widely chosen method to solve MOP.

From Table 12, it can be observed that there are 26 exact algorithms proposed in multi-objective GVRP studies. Meanwhile, the number of multi-objective GVRP studies employing this approach is 22, as indicated in Fig. 13. Notably, one researcher, Delfani et al. [395], proposed multiple exact algorithms. They presented a multi-objective optimization model for a hazardous waste location-routing problem and introduced five exact algorithms: lap-metric, goal attainment, max-min, goal programming, and weighted sum to solve the MOP model.

Table 12 presents the prevalent utilization of the ε-constraint method in multi-objective GVRP studies. Ghezavati and Beigi [383] employed the ε-constraint method to address a bi-objective model using the Baron solver in the GAMS software for a small test problem. Additionally, they introduced a metaheuristic approach based on the non-dominated sorting genetic algorithm II (NSGA-II) for both small and large test problems. Ebrahimi [32] developed a stochastic multi-objective MILP model integrating supplier selection with location-allocation-routing problems. They applied an augmented ε-constraint method as a multi-objective decision-making tool to solve the proposed model.

Table 12
Exact algorithms proposed to solve multi-objective GVRP.

Exact algorithm	Authors	Quantity
Chance-constraint and-/Epsilon constraint	Zarouk et al. [382]/Ebrahimi [32], Nosrati and Khamseh [69], Wang et al. [87], Ghezavati and Beigi [383], Ouhader and El Kyal [384], Kabadurmuş et al. [385], Abdullahi et al. [386], Tirkolaee et al. [387], Talouki et al. [388], Heidari et al. [389], Roosta et al. [390], Santos et al. [391]	13
Weighted-/Revised multi-choice-/ Branch and cut in fuzzy-/Goal programming	Hashemi-Amiri et al. [392]/ Kordi et al. [393]/ Peivastehgar et al. [394]/Delfani et al. [395]	4
Weighted sum	Delfani et al. [395], Sawik [396], Wu and Dong [397]	3
Lp-metric method	De and Giri [50], Delfani et al. [395]	2
Weighted-/Max-min method	Govindan et al. [398]/Delfani et al. [395]	2
Cutting most of the non-Pareto-optimal solutions	Yu et al. [399]	1
Goal attainment method	Delfani et al. [395]	1
Total exact algorithms proposed for multi-objective GVRP		26

Futhermore, Nosrati and Khamseh [69] investigated a bi-objective hybrid vehicle routing problem, formulating the model as mixed integer nonlinear programming, and employing multi-objective simulated annealing algorithm (MOSA) algorithm & ε-constraint method. The weighted sum method emerges as the second most frequently adopted exact algorithm by researchers for addressing multi-objective GVRP from 2016 to 2023. Sawik [396] proposed the weighted sum method for bi-objective GVRP with the aim of simultaneously maximizing truck capacity while minimizing fuel usage, carbon dioxide emissions, and noise reduction.

6.7. Heuristic algorithm for solving multi-objective GVRP

Table 13 presents four researchers who proposed a heuristic algorithm to solve multi-objective GVRP. Birecik and Özgen [400] proposed an interactive fuzzy method to tackle the green capacitated vehicle routing problem with uncertain travel times for both vehicles and supplier demands. They modeled uncertainty using triangular fuzzy numbers and formulated the optimization problem as a bi-objective possibilistic mixed-integer programming (PMIP) model.

6.8. Metaheuristic algorithm for solving multi-objective GVRP

Table 14 shows that there are 98 metaheuristic algorithms proposed to solve multi-objective GVRP studies from 2016 to 2023. 81 out of 98 proposed metaheuristic algorithms belonging to the classification of P-metaheuristics. NSGA-II and its variants emerged as the most frequently proposed P-metaheuristic algorithms for multi-objective GVRP, followed by PSO along with their variants. On the other hand, S-metaheuristic algorithms were only proposed in 17 studies, with SA being the most used algorithm for solving multi-objective GVRP. According to Table 14, evolutionary multi-objective optimization (EMO) algorithms are also widely used in solving multi-objective GVRP. Out of the 81 studies proposing P-metaheuristic algorithms, 55 of them proposed algorithms belonging to the EMO category. One of the advantages of EMO is its capability to discover a set of Pareto solutions in a single run within a reasonable time [404]. Futhermore, genetic algorithm-based solution approaches within evolutionary algorithms (EAs) are increasingly recognized for their effectiveness in addressing multi-objective optimization [405]. Based on the strengths of GA, there are 45 out of 55 studies proposing EMO algorithms based on GA to solve multi-objective GVRP. Some algorithms introduced by GVRP researchers include non-dominated sorting genetic algorithm II (NSGA-II), NSGA-III, multi-objective genetic algorithm (MOGA), and non-dominated ranking genetic algorithm (NRGA). The metaheuristic algorithms proposed in a single study of multi-objective GVRP can be found in Appendix 4.

Table 13
Heuristic algorithms proposed to solve multi-objective GVRP.

Heuristic algorithm	Authors	Quantity
Interactive possibilistic programming approach	Birecik and Özgen [400]	1
K-shortest path algorithm	Kumar et al. [401]	1
Algorithm name is not written explicitly	Mohammadi et al. [402], Giovannelli and Vecente [403]	2
Total heuristic algorithms proposed for multi-objective GVRP		4

Deb et al. [457] proposed the NSGA-II as a more elitist and efficient version of the previous NSGA, having few parameters and is computationally faster. To guarantee population diversity, the NSGA-II algorithm first generates a population of individuals, then ranks, and sorts each one according to the nondomination level, and finally performs crowding distance sorting. It reduces the complexity of noninferior ranking genetic algorithms, which gives the advantage of fast operation speed and good solution set convergence. Due to its advantages, NSGA-II emerges as the predominantly proposed algorithm for solving MO-GVRP. As depicted in Table 14, among the 55 studies proposing EMO algorithms, 35 of them endorse NSGA-II.

Several researchers also made modifications and improvements to the NSGA-II to improve the convergence of the algorithm. Asghari and Mirzapour Al-e-hashem [109] proposed a self-learning NSGA-II to solve the Green Delivery-Pickup Problem for home hemodialysis machines. The proposed algorithm is inspired by the self-learning genetic algorithm proposed by Kostenko and Frolov [458] to solve the single-objective optimization problem. Bai et al. [101] proposed low-carbon vehicle routing optimization model for cold chain logistics that considers real-time traffic conditions in the actual road network. To further increase the convergence of the population and the precision of the search direction, they proposed an improved NSGA-II method based on adaptive crossover and mutation probabilities.

Swarm Intelligence (SI) algorithms have also gained popularity in recent years. PSO and ACO are the most widely proposed algorithms to solve MO-GVRP compared to other SI algorithms, as demonstrated in Table 15. Various multi-objective swarm intelligence (MOSI) algorithms have been developed to handle multi-objective optimization problem (MOP). In practice, MOSI algorithm combines a single-objective swarm intelligence algorithm with a multi-objective optimization approach [459]. Li et al. [432] developed a multi-depot green vehicle routing problem that maximizes revenue while minimizing costs, time, and emissions. They improved the conventional ACO algorithm by proposing a new technique to update the development of pheromones in solving the model. The performance test results of the Improved ACO show satisfying performance when compared to the optimal solutions produced by LINGO and it outperformed the conventional ACO algorithm. Rabbani et al. [409] applied NSGA-II and MOPSO to solve the transportation-location-routing problem with soft time windows. Based on the obtained results, MOPSO algorithm performs better than the NSGA-II algorithm in the number of Pareto front solutions (NPS), spacing metric (SM), and diversification of solutions in the space (DM), although NSGA-II generally has a shorter computational time than MOPSO.

Several novel SI algorithms have been introduced by researchers, including the multi-objective grey wolf optimizer (MOGWO), modified salp swarm algorithm (MSSA), red deer algorithm (RDA), influenza virus algorithm (IVA), and membrane-inspired multi-objective algorithm (MIMOA). Abad et al. [106] proposed the multi-objective imperialist competitive algorithm (MOICA) and multi-objective grey wolf optimizer (MOGWO) to address the split pollution routing problem with controlled indoor activities in cross-docking under uncertainty. Their assessment, conducted across 30 test problems utilizing five performance metrics,

Table 14
Metaheuristic algorithms proposed to solve multi-objective GVRP.

Metaheuristic algorithm	Category	Authors	Quantity
Multifactorial evolutionary algorithm-/Improved-/Self learning-/ /Extended-/NSGA-II/-teaching learning-based optimization	P	Yin [406]/Xu et al. [88], Bai et al. [101]/ Asghari and Mirzapour Al-e-hashem [109]/ Rauniyar et al. [407]/Zhang et al. [408]/Alinaghain and Zamani [76], Rabbani et al. [77], Wang et al. [87], Rabbani et al. [89], Abad et al. [105], Asghari and Mirzapour Al-e-hashem [109], Leng et al. [113], Ghezavati and Beigi [383], Rabbani et al. [409], Jemai et al. [410], Ghannadpour [411], Rahimi et al. [412], Asadi et al. [413], Zapata-cortes [414]/Ganji et al. [39], Tirkolaee et al. [387], Heidari et al. [389], Roosta et al. [390], Amiri et al. [392], Giallanza and Puma [415], Tabatabaei et al. [416], Alamatsaz et al. [417], Bavar et al. [418], Hajghani et al. [419], Kuo et al. [420], Abadi et al. [421], Guo et al. [422], Li et al. [423]/Zahedi and Kia [424]	35
Improved-/MOPSO/ Multi-objective self learning PSO	P	Kuo et al. [425]/ Ganji et al. [39], Rabbani et al. [89], Abad et al. [105], Rabbani et al. [409], Asadi et al. [413]/Kumar et al. [426]	7
Modified-/Discrete multi-swarm-/PSO with greedy mutation operator	P	Norouzi et al. [427] /Rau et al. [428]/Poonthalir and Nadarajan [429]	3
Modified-/Cycle evolutionary-/Rule-based fuzzy-/GA	P	Ghannadpour and Zarrabi [46]/ Qin et al. [430]/ Gupta et al. [431]/Rezaei et al. [31]	4
Improved-/MO-/ACO	P	Li et al. [432], Wang et al. [433]/ Ganji et al. [39]/Zohal and Soleimani [434]	4
Improved-/NSGA-III	P	Zhou et al. [435]/Leng et al. [113], Zabihian-Bisheh et al. [436]	3
Improved-/Modified differential evolution	P	Fallah et al. [437]/Kumar et al. [438]	2
Strength Pareto evolutionary algorithm II	P	Leng et al. [113], Jemai et al. [410]	2
Indicator-based evolutionary algorithm (IBEA)	P	Leng et al. [113], Jemai et al. [410]	2
Multi-objective grey wolf optimizer (MOGWO)	P	Abad et al. [106], Heidari et al. [389]	2
Other P-metaheuristic algorithms	P	Abad et al. [105], Abad et al. [106], Abdi et al. [107], Leng et al. [113], Hajghani et al. [419], Mojtahedi et al. [439], Cai et al. [440], Psychas et al. [441], Zulvia et al. [442], Niu et al. [443], Fathollahi-Fard et al. [444], Validi and Bhattacharya [445], Lu et al. [446], Li et al. [447], Gupta et al. [448], Coelho et al. [449], Psychas et al. [450]	17
Total P-metaheuristic algorithms proposed for multi-objective GVRP			81
Population-based-/Modified-/Multi-objective-/Archived multi-objective- /Simulated annealing	S	Rezaei et al. [31]/Fathollahi-Fard et al. [444]/Nosrati and Khamseh [69], Tirkolaee et al. [387], Abadi et al. [421]/Heilig et al. [451]/ Rabbani et al. [97], Abdi et al. [107], Mohammadbagher and Torabi [452]	9
Enhanced-/Adaptive large neighborhood search	S	Tan et al. [453]/Wen et al. [454], Santos et al. [391]	3
Other S-metaheuristic algorithms	S	Eskandarpour et al. [45], Leng et al. [113], Londono et al. [123], Xia et al. [455], Cao et al. [456]	5
Total S-metaheuristic algorithms proposed for multi-objective GVRP			17
Total metaheuristic algorithms proposed for multi-objective GVRP			98

demonstrated the superior performance of MOGWO compared to MOICA. Additionally, Fathollahi-Fard et al. [444] proposed a new mathematical formulation for the home health care routing problem, considering environmental pollution, and introduced modified salp swarm algorithm (MSSA) and modified simulated annealing (MSA) as solution methodologies for this problem.

6.9. Hyper-heuristic algorithm for solving multi-objective GVRP

Hyper-heuristics for multi-objective optimization problems represent a novel research domain to solve GVRPs. Multi-objective hyper-heuristics (MOHH) refer to a search method or learning process aimed at addressing multi-objective optimization problems by effectively integrating and managing the capabilities of a predetermined set of low-level heuristics [461]. Researchers such as Leng et al. [460], Leng et al. [461], and Zhang and Zhang [462] have proposed hyper-heuristic algorithms tailored to address the multi-objective green vehicle routing problem (MO-GVRP). Leng et al. [460] introduced a MOHH technique for solving the regional low carbon location-routing problem, wherein high-level heuristics (HLH) were utilized, incorporating four selection techniques and three acceptance criteria, alongside three multi-objective evolutionary algorithms serving as low-level heuristics. Furthermore, Leng et al. [461] presented a MOHH approach employing decomposition strategies (MOHH/D) to tackle the low-carbon cold chain location-routing problem.

6.10. Hybrid algorithm for solving multi-objective GVRP

As indicated in Table 15, five works have introduced hybrid exact-metaheuristics for multi-objective GVRP. Amiri et al. [392] and Amiri et al. [463] contributed two algorithms, resulting in a total of seven hybrid exact-metaheuristic algorithms. Anderluh et al. [464] combined large neighborhood search (LNS) with the ε-constraint method to approximate the set of Pareto-optimal solutions. Meanwhile, Zhou et al. [465] proposed a two-phase approach for solving the fuzzy chance-constraint programming model considering the trade-off between the total cost and the recycling revenue. In the first phase, the model is converted into a deterministic nonlinear form, and in the second phase, a genetic algorithm (GA) is employed to solve the equivalent model.

Table 16 presents the combination of exact and heuristic algorithms proposed by researchers for multi-objective GVRP. The exact methods employed include branch-and-bound, augmented ε-constraint, and weighted sum methods. Research conducted by [96,467], and [468] employed branch-and-bound method. Tricoire and Parragh [467] developed a decomposition strategy, constructing routes and utilizing them in a separate set covering model to produce comprehensive solutions. They employed bi-objective branch-and-bound (BIOBAB) to tackle the model and applied 2-opt improvement in the generated routes. Daşcıoğlu and Yazgan [468] proposed a cluster-first-route-second methodology. In the clustering phase, the location problem was addressed using the green p-median algorithm. Subsequently, the routing problem was formulated as a MILP model and solved using the branch-and-bound method.

During the period of 2016–2023, our review revealed that only one GVRP study employed a combination of two heuristic methods to address the multi-objective GVRP. Abdullahi et al. [386] introduced a biased-randomised iterated greedy with local search (BRIG-LS) method to tackle the sustainable VRP. The BRIG-LS algorithm integrates a biased-randomised approach with the iterated greedy algorithm, which incorporates a local search technique utilizing random swaps.

Table 17 presents the combination of metaheuristics with complementary metaheuristics proposed for addressing multi-objective GVRP. According to the review findings, this combination type is more

Table 15
Hybrid exact-metaheuristic algorithms proposed for multi-objective GVRP.

Metaheuristic algorithm	Exact algorithm			Total
	ε-constraint	Weighted sum	Chance-constraint programming	
A-/LNS	[392,463, 464]	[392,463]		5
GA			[465]	1
GRASP-ELS	[466]			1
Total	4	2	1	7

GRASP-ELS: Greedy randomized adaptive search procedure-evolutionary local search.

prevalent compared to combining metaheuristic algorithms with exact methods, heuristic algorithms, or simulations. As shown in Table 17, there are 22 hybrid metaheuristic-metaheuristic algorithms proposed by researchers focusing on multi-objective GVRP. Among 20 researchers proposing hybrid metaheuristic-metaheuristic, Abdi et al. [107] and Amiri et al. [463] each proposed 2 algorithms. Abdi et al. [107] used GA algorithm for generating an initial solution and employed keshtel algorithm (KA) and PSO algorithms during the mutation stage of the GA algorithm to achieve better solutions.

From Table 17, NSGA-II and GA algorithms remain widely used for solving multi-objective GVRP. S-metaheuristics such as tabu search and VNS, known for their effectiveness in exploiting Pareto sets, have been embedded with the multi-objective P-metaheuristics (NSGA-II, GA). Chen et al. [470] introduced a novel hybrid multi-objective approach named NSGA-II-TS, which integrates the TS method into the NSGA-II framework. In each generation, some chromosomes in the current population are selected based on specific criteria, and the TS procedure is then applied to them. This inclusion of the TS procedure enhances solution diversity, thereby expectedly improving the quality of the nondominated solution set. Additionally, another study [471] proposed a combination of NSGA-II and TS, employing a different hybridization procedure.

According to Table 18, multi-objective GVRP researchers have proposed 21 hybrid heuristic-metaheuristic algorithms. Out of the 19 studies utilizing the hybrid heuristic-metaheuristic approach as indicated in Fig. 13, studies by [471] and [483] each proposed 2 algorithms, bringing the total number of algorithms to 21. Within Table 18, NSGA-II was utilized in 7 studies, establishing it as the predominant algorithm paired with heuristic algorithms for addressing multi-objective GVRP.

From the results of the review, only two studies proposed a hybrid simulation-optimization algorithm for multi-objective GVRP. Wang et al. [499] presented a green-fuzzy VRP modelled with the credibility chance constraint and proposed an optimized hybrid intelligent algorithm along with an enhanced fuzzy simulation. Zhang et al. [500] proposed an uncertain multi-objective location-routing programming for emergency response with consideration of travel time, emergency relief costs and carbon dioxide emissions. They developed a hybrid intelligent algorithm combining uncertain simulation and a genetic algorithm to tackle the proposed model.

Table 16
Hybrid exact-heuristic algorithms proposed for multi-objective GVRP.

Heuristic algorithm	Exact algorithm			Total
	Branch-and-bound	Augmented ε-constraint	Weighted sum	
2-opt algorithm	[467]			1
Heuristic method		[388]		1
Neighborhood search			[469]	1
Path elimination procedure algorithm	[96]			1
p-median algorithm	[468]			1
Total	3	1	1	5

Table 19 provides a consolidated summary based on the metaheuristic algorithms frequently in Tables 14-15 and 17-18. Among these algorithms, NSGA-II, SA, and PSO, along with their variants, emerge as the top choices proposed by researchers for multi-objective GVRP studies. It's noteworthy that NSGA-II stands out as the algorithm most prevalently used by researchers in addressing multi-objective GVRP. This alligns with the findings of the review by [6]. Furthermore, the review findings by [5] also indicate that GA and PSO are predominant algorithms in MO-GVRP.

7. Conclusions

This study presents a literature review that focuses on solution approaches and algorithms for green vehicle routing problem and its variants. Through a systematically searched strategy, 458 papers were identified and reviewed. The results of the review related to the characteristics of the GVRP indicate that the majority of researchers consider Green-VRP category, homogeneous fleet of vehicle, fuel type is gasoline or diesel, time window constraints, and single-objective optimization with total cost including emission cost as the objective function. Out of the 458 papers analyzed, 424 put forth a mathematical model to tackle the GVRP. Subsequently, 229 researchers developed MM and implemented their proposed solution approach. The findings from the review indicate that hybrid and metaheuristic methods are the predominant solution approaches for single-objective GVRP. Among various hybrid types, the combination of metaheuristics-metaheuristics is the preferred choice among GVRP researchers. Additionally, large neighborhood search (LNS) and its variants (especially adaptive large neighborhood search) is identified as the most commonly used algorithm in single-objective GVRP. This algorithm is proposed in both metaheuristic solution approaches and hybrid approaches. In contrast, metaheuristics emerge as the dominant solution approach for multi-objective GVRP, with NSGA-II being the most frequently used algorithm for addressing multi-objective GVRP.

From this review study, we identify several challenges in the green vehicle routing problem (GVRP) and propose potential directions for future research:

1. In many previous studies of GVRP, several uncertainties such as customer demand fluctuations, traffic congestion, and unpredictable weather patterns have frequently been disregarded. It is imperative for future research initiatives to integrate these uncertainties into GVRP models and develop various approaches, encompassing fuzzy logic, stochastic modeling, and robust optimization techniques.
2. The results indicate that single-objective optimization has emerged as the prevailing approach in tackling the green vehicle routing problem. Therefore, the challenge for future research lies in developing multi-objective optimization to address this problem more comprehensively.
3. In light of the growing focus on sustainability, optimizing vehicle routing to reduce fuel consumption and carbon emissions has become essential. In multi-objective GVRP, there should be an integration of environmental objectives with other social objectives such as social impact, transportation risk, and workload balancing.
4. Although evolutionary algorithms are widely utilized in GVRP research, swarm intelligence (SI) algorithms have not received as much attention. Investigating the capabilities of SI algorithms, which have demonstrated success in optimizing traditional vehicle routing problems, holds significant potential for enhancing GVRP studies.
5. The exploration of sophisticated optimization methods and algorithms is crucial for research on both single and multi-objective GVRP. Utilizing a hybrid approach can effectively address these challenges. According to the review findings, hybrid exact-metaheuristics and hybrid metaheuristics-metaheuristics show promise for further development in this regard.

Table 17
Hybrid metaheuristic-metaheuristic algorithms proposed for multi-objective GVRP.

Metahe-uristics algorithm	TS	PSO	VNS	ALNS	HACO	SA	BEA	CA	KA	SSA	Total
NSGA-II	[470,471]		[472,473,474]	[475]		[436]	[108]				8
GA	[476]	[99,107]			[477]	[382]		[478]	[107]		7
SA	[479]	[480]		[463]						[444]	4
ALNS	[463]										1
ABCA					[481]						1
PAES	[482]										1
Total	6	3	3	2	2	2	1	1	1	1	22

BEA: Bee evolutionary algorithm, KA: Keshtel algorithm, HACO: Hybrid ant colony optimization, CA: Cultural algorithm, ABCA: Artificial bee colony algorithm, PAES: Pareto archived evolution strategy.

Table 18
Hybrid heuristic-metaheuristic algorithms proposed for multi-objective GVRP.

Metahe-uristic	Heuristic algorithm												To-
Algo-rithm	CW	KCS	LS	Gr	NN/-C	GSC	KM	PHA	RLSH	RS	RSH	RTH	tal
NSGA-II	[114,471,484]	[485,486]	[487]	[488]									7
A-/LNS			[489]							[464]			2
GA					[490]			[417]					2
MOGA				[483]		[483]							2
MOPSO	[491]		[492]										2
SPEA/ SPEA2		[493]							[494]				2
ABC												[495]	1
ACS											[496]		1
ILS					[497]								1
NSGA-II & TS	[471]												1
NSGA-III							[498]						1
Total	5	3	3	2	2	1	1	1	1	1	1	1	21

RS: Rectale splitting, RTH: Route truncation heuristic, CW: Clarke and wright's savings, KCS: K-means clustering algorithm, GSC: Global search cluster, PHA: Progressive hedging algorithm, RSH: Routing and scheduling heuristic, RLSH: Resultant local search heuristic, SPEA: Strength pareto evolutionary algorithm, NNC: nearest-neighbor algorithm based on minimum cost.

Table 19
Summary based on metaheuristic algorithms frequently employed in solving multi-objective GVRP.

Metaheuristic algorithm	Metaheuristic category	Most frequently proposed algorithms based on the table				Total
		14	15	17	18	
NSGA-II and its variants	**P**	35	–	8	9	52
SA and its variants	S	9	–	6	–	15
PSO, MOPSO and its variants	**P**	10	–	3	2	15
GA and its variants	**P**	4	1	7	2	14
A-/LNS	S	3	5	3	2	13
TS	S	1	–	6	1	8
ACO	**P**	4	–	1	1	6

6. The development of real-time routing algorithms capable of adjusting to dynamic factors such as traffic, weather, and customer requirements is becoming increasingly prevalent. Simulation-optimization methods, particularly simheuristics, have emerged as a primary approach for generating high-quality routing plans in uncertain conditions.
7. The combination of metaheuristics with data mining or machine learning presents an opportunity to enhance the effectiveness and efficiency of optimization processes. However, this hybrid combination has received limited attention from researchers in the field of green vehicle routing problem (GVRP). Furthermore, it remains relatively underexplored as a hybrid approach type in the categorization of review papers. These findings suggest that future research should delve further into exploring and advancing hybrid metaheuristics with data mining or machine learning to address complex models within the field of GVRP, particularly in the context of multi-objective studies.
8. Understanding the combined algorithms utilized in hybrid approaches provides guidance for future research directions. Researchers can expand upon existing hybrid approaches or investigate novel combinations to enhance solution quality, efficiency, and suitability for real-world green vehicle routing problems. Therefore, studies analyzing in greater detail the main strategies or operations of hybrid solution methods are crucial to be conducted in the future.
9. Although hyper-heuristics have shown effectiveness in tackling various combinatorial optimization problems, their adoption in GVRP is comparatively limited. It is crucial to conduct further research and exploration into the application of hyper-heuristics within the context of GVRP to promote their wider utilization.

CRediT authorship contribution statement

Annisa Kesy Garside: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Robiah Ahmad:** Writing – review & editing, Validation, Supervision. **Mohd Nabil Bin Muhtazaruddin:** Writing – review & editing, Writing – original draft, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix

Appendix 1, Appendix 2, Appendix 3, Appendix 4

Appendix 1
Exact algorithms proposed by only one study in single-objective GVRP.

Exact algorithm	Authors	Quantity
An exact algorithm based on the set-partitioning model	Yu et al. [196]	1
An exact method for solving set-partitioning (set partitioning heuristic, dynamic programming algorithm, column generation method)	Andelmin and Bartolini [185]	1
Exact algorithms to obtain the POS_C and POS_CE	Wang et al. [186]	1
An exact algorithm with three steps	Sun et al. [30]	1
An iterated beam search	Yavuz [187]	1
Branch and bound	López-Castro et al. [188]	1
Cumulative location-routing and speed optimization algorithm	Dukkanci et al. [189]	1
Disjunctive convex programming	Fukusawa et al. [190]	1
Exact two-phase decomposition method	Liu et al. [191]	1
Fuzzy bipolar approach	Shirzadi et al. [192]	1
Multiple origin-destination, combinatorial upper confidence bound, covariance link-level energy consumption, path Elimination	Chen et al. [193]	1
Path-based exact solution approach	Bruglieri et al. [194]	1
Unified branch-and-Benders-cut	Maheo et al. [195]	1

Appendix 2
Heuristic algorithms proposed by only one study in single-objective GVRP.

Heuristic algorithm	Authors	Quantity
Clarke and wright heuristic & combination of several swap moves	Wehbi et al. [501]	1
Cost change estimates heuristic	Saka et al. [200]	1
Exact solution heuristic	Saka et al. [200]	1
Hybrid heuristic	Saka et al. [200]	1
Heuristic 1: adaptive recharging only	Sweda et al. [197]	1
Heuristic 2: adaptive routing and recharging	Sweda et al. [197]	1
GROS-I	Alinaghian and Zamani [35]	1
GROS-II	Alinaghian and Zamani [35]	1
GGT	Alinaghian and Zamani [35]	1
Alternating direction method of multipliers	Yang et al. [317]	1
Cluster first route second	Ding et al. [502]	1
Fusion heuristic algorithm	Zhang et al. [503]	1
Greedy algorithm	Solano et al. [199]	1
Lagrangian relaxation algorithm	Ren et al. [290]	1
Maximum waste in a minimum time during each trip (MWMTT)	Kaabi et al. [504]	1
Multi-space sampling heuristic	Montoya et al. [58]	1
Multi-start biased-randomized heuristic	Almouhanna et al. [201]	1
Novel multi-agent reinforcement learning	Wang et al. [505]	1
Path based heuristic approach	Bruglieri et al. [194]	1
Safe reinforcement learning	Basso et al. [506]	1
The adapted hierarchical agglomerative clustering	Köseli et al. [507]	1
The approximate dynamic programming-based heuristic algorithm	Çimen and Soysal [103]	1
Transformer-based deep reinforcement learning method	Tang et al. [508]	1
Two-phase decomposition MIP based heuristic	Chekoubi et al. [509]	1
Total heuristic algorithms proposed for single-objective GVRP		24

Appendix 3
Metaheuristic algorithms proposed by only one study in single-objective GVRP.

Metaheuristic algorithm	Category	Authors	Quantity
Bee colony optimization	**P**	Stamadianos et al. [216]	1
Clustering-based artificial immune algorithm	**P**	Liu et al. [244]	1
Flower pollination	**P**	Pratap et al. [241]	1
Intelligent water drop	**P**	Chen et al. [245]	1
Modified camel	**P**	Utama et al. [246]	1
Modified differential search	**P**	Quan et al. [247]	1

(*continued on next page*)

Appendix 3 (*continued*)

Metaheuristic algorithm	Category	Authors	Quantity
Multi-round iterated greedy	**P**	Hatami et al. [36]	1
Social engineering optimizer	**P**	Rahmanifar et al. [249]	1
Type 1 and type-2 fuzzy logic discrete fireworks	**P**	Thakur et al. [248]	1
Greedy randomized adaptive search procedure (GRASP)	S	Bruglieri et al. [291]	1
Large neighborhood search algorithm based on Q-learning	S	Ren et al. [290]	1

Appendix 4

Metaheuristic algorithms proposed by only one study in multi-objective GVRP.

Metaheuristic algorithm	Category	Authors	Quantity
Adaptive memory social engineering optimizer (AMSEO)	**P**	Mojtahedi et al. [439]	1
Artificial physarum polycephalum colony	**P**	Cai et al. [440]	1
Grid-based evolutionary algorithm (GrEA)	**P**	Leng et al. [113]	1
Influenza virus algorithm (IVA)	**P**	Psychas et al. [441]	1
Many-objective gradient evolution (MOGE)	**P**	Zulvia et al. [442]	1
Membrane-inspired multi-objective algorithm (MIMOA)	**P**	Niu et al. [443]	1
Modified salp swarm algorithm (MSSA)	**P**	Fathollahi-Fard et al. [444]	1
MOGA-II	**P**	Validi and Bhattacharya [445]	1
Multi-objective brainstorming optimization (MOBSO)	**P**	Lu et al. [446]	1
Multi-objective imperialist competitive algorithm (MOICA)	**P**	Abad et al. [106]	1
Multi-objective immune genetic algorithm (MOIGA)	**P**	Li et al. [447]	1
Multi-objective shortest path evolutionary algorithm (MOSPEA)	**P**	Gupta et al. [448]	1
Multi-objective smart pool search (MOSPOOLS)	**P**	Coelho et al. [449]	1
NRGA	**P**	Abad et al. [105]	1
Parallel multi-start non-dominated sorting differential evolution algorithms (PMS-NSDEs)	**P**	Psychas et al. [450]	1
Red deer algorithm (RDA)	**P**	Abdi et al. [107]	1
Multi-objective stochastic fractal search (MOSFS)	**P**	Hajghani et al. [419]	1
Adaptive TS	S	Xia et al. [455]	1
Enhanced multi-directional local search (EMDLS)	S	Eskandarpour et al. [45]	1
Iterated local search (ILS)	S	Londono et al. [123]	1
Memory-based variable neighborhood search	S	Cao et al. [456]	1
Nondominated sorting and local search (NSLS)	S	Leng et al. [113]	1

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